



Faculty of Business Management Sciences and Economics

**SUPERVISED MACHINE LEARNING PREDICTIVE MODEL FOR
PERSONALIZED MEDICAL INSURANCE PREMIUMS.**

A CASE STUDY OF CIMUS MEDICAL AID SOCIETIES

BY

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Declaration

I, Rudo CHIYAKA, declare that this dissertation entitled:

“Machine Learning Model for Personalized Premium Pricing in the Health Insurance Sector:
A Case Study of CIMAS Medical Aid Society in Zimbabwe”

is my original work and has not been submitted in full or in part for any degree or diploma at any other university or institution of higher learning. All sources used have been duly acknowledged through appropriate referencing, and this dissertation is submitted in partial fulfillment of the requirements for the award of the Honors Degree in Marketing Informatics at University of Zimbabwe.

I am fully aware of the consequences of plagiarism and affirm that this work complies with the university’s academic integrity and ethical research guidelines.

Signature: _____

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Dedication

This work is lovingly dedicated to my mother, whose unwavering love, prayers, and strength have been the foundation of everything I am. To my sister Prudence, thank you for your constant support, encouragement, and belief in me even when the journey became tough. To my precious two-year-old son Tanaka, born during this academic chapter of my life you are my greatest inspiration. Your presence gave me the strength to push harder and strive for excellence; I dedicate this achievement to the future I dream of for you. Finally, to my dear friends, thank you for walking with me through the highs and lows of this journey your patience, motivation, and faith in me have meant the world.

Abstract

The rising cost of healthcare in Zimbabwe has revealed the deficiency of traditional actuarial method of medical insurance premium pricing, based on high dependency on demographic data such as gender and age. This study looks at the application of machine learning (ML) techniques to develop a predictive model of tailored medical insurance premiums, and CIMAS Medical Aid Society was utilized as an example. The primary objective was to create a machine learning based model that provides precise predictions of individual healthcare costs based on health risk factors and utilization behaviour, thereby enabling sustainable and equitable premium pricing. Adopting a quantitative research approach and the CRISP-DM process, the study employed supervised ML algorithms, including Linear Regression, Gradient Boosting, Artificial Neural Networks (ANN), Random Forest and XGBoost. Regression to analyse controlled data from CIMAS. Feature engineering, cross-validation, and hyperparameter optimization were employed to optimize model performance. Among all the algorithms attempted, Random Forest was the most robust, having an R^2 of 0.9563 and an RMSE of 0.21, indicating excellent predictive accuracy. The study indicated that acute indicators of healthcare use, such as emergency room use, chronic drug use, and heart disease, were the most significant predictors of healthcare expenditure. In contrast, lifestyle factors like smoking and BMI had little impact on cost modelling. The predictive model was subsequently used to model premium mapping, which revealed inconsistencies between current plan assignments and risk-based recommendations, and thus the inefficiencies of fixed pricing models. According to the results, the research suggests that insurers adopt AI-driven pricing models and employ explainable AI tools (e.g., SHAP values) to enable transparency and fairness. It also advocates for integrating real-time health data, such as wearables and electronic health records, to enhance model responsiveness. The research concludes that ML-based personalized premium models can significantly improve fairness, affordability, and efficiency in Zimbabwe's health insurance sector.

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Abbreviations and Acronyms ITEM

Abbreviation	Full Term
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
ANN	Artificial Neural Network
SVM	Support Vector Machine
KNN	K-Nearest Neighbors
EHR	Electronic Health Record
CDSS	Clinical Decision Support System
R^2	Coefficient of Determination
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MSE	Mean Squared Error
PCA	Principal Component Analysis
CIMAS	Commercial and Industrial Medical Aid Society
CSV	Comma-Separated Values
CRISP-DM	Cross-Industry Standard Process for Data Mining
BMI	Body Mass Index
VIF	Variance Inflation Factor
ER	Emergency Room
OP	Outpatient
IP	Inpatient
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
EUT	Expected Utility Theory
API	Application Programming Interface
GPU	Graphics Processing Unit

CHAPTER ONE: INTRODUCTION

1.0 Introduction

This study focuses on designing a supervised machine learning predictive model designed to generate personalized insurance premiums based on individual health profiles and associated healthcare cost, specifically within the framework of CIMAS medical aid in Zimbabwe. In this first chapter, the study provides a comprehensive overview that begins with the background, setting the stage for the necessity of personalized premium pricing in the health insurance landscape. Following this, the problem statement outlines the challenges inherent in current pricing models. The chapter further presents the conceptual framework that illustrates the relationships among key variables, along with the objectives and research questions that guide the investigation. Additionally, it discusses the significance of the study, highlighting its potential benefits for various stakeholders, and concludes by delineating the delimitations that define the study's scope and boundaries.

1.1 Background of the study

Machine Learning

Machine learning is a key subset of artificial intelligence, involves the development of algorithms that enable systems to identify patterns in large datasets, adaptively improve performance, and make autonomous decisions without being explicitly programmed for every task. In the context of healthcare, ML models are particularly valuable for predicting outcomes such as treatment costs by learning from historical patient and claims data. Common ML approaches include supervised learning (which uses labelled datasets), unsupervised learning (which detects hidden patterns in unlabeled data), and reinforcement learning (which optimizes decision-making through feedback loops) (Usman *et al.*, 2023; Zhang *et al.*, 2022). These techniques are increasingly being applied to enhance cost estimation and resource planning in health systems (Kourou *et al.*, 2021; Bhavani Prasad *et al.*, 2022).

Supervised learning

Supervised learning is a fundamental machine learning approach that involves training models using labelled data, where the input features are mapped to known outcomes. During this process, the algorithm learns the relationships between variables so that it can accurately predict outcomes when presented with new, unseen data. This method is widely used for both classification and regression tasks. Popular supervised learning algorithms include linear regression, logistic regression, k-nearest neighbours (KNN), support vector machines (SVMs),

decision trees, random forests, gradient boosting methods, and artificial neural networks (ANNs) (Zhang *et al.*, 2022; Alzubaidi *et al.*, 2021; Reddy *et al.*, 2022). These models have proven effective in healthcare for predicting medical costs, disease risks, and treatment outcomes.

Deep Learning (DL)

Deep learning (DL), a specialized branch of machine learning, leverages artificial neural networks composed of multiple hidden layers to model complex relationships within large and high-dimensional datasets. In healthcare, DL techniques particularly deep neural networks are capable of processing unstructured data such as clinical notes, diagnostic images, and electronic health records. This capability allows DL models to detect subtle patterns and correlations that conventional machine learning methods may overlook. As a result, these models support the creation of more nuanced and holistic health risk profiles, incorporating both diagnosed medical conditions and potential future risks based on trends identified in the data (Xiong *et al.*, 2023; Esteva *et al.*, 2020; Lee & Kim, 2022).

Medical Aid Plans

Medical aid plans are integral components of the healthcare system, providing essential financial coverage for various medical services, including hospitalization, outpatient care, and preventive treatments. These plans are designed to alleviate the financial burden of healthcare expenses for individuals and families, ensuring access to necessary medical services. Key features of medical aid plans include premium fees, maximum coverage limits, co-payments, and exclusions. Premiums represent the regular payments made by policyholders to maintain their coverage, while coverage limits define the maximum amount an insurer pays for specific services during a designated period. However, the complexity of medical aid plans can often lead to confusion and dissatisfaction among policyholders, particularly when they encounter high costs or intricate policy language. This underscores the need for a more transparent and equitable approach to premium pricing that takes individual health risks into account (Albahri *et al.*, 2023).

Personalised Medical premiums

Personalized health plans offer a customized approach to well-being by addressing a person's individual health requirements, desires, and daily habits (Smith & Johnson, 2022). Personalized medical premiums are a transformative innovation in health insurance, enabled

by the predictive power of advanced machine learning models. Traditional pricing models, which rely on basic demographics like age and gender, often fail to grasp the nuanced interaction of individual health risks and healthcare utilization patterns (Kasaraneni (2021) By contrast, machine learning techniques such as gradient boosting, deep neural networks, and recurrent neural networks allow insurers to analyse vast datasets including electronic health records, lifestyle, and pharmacy claims data. These models can uncover non-linear relationships and synergistic effects of comorbidities something traditional models typically overlook. As a result, insurers can more accurately estimate an individual's expected healthcare costs and tailor premiums accordingly, promoting both fairness and efficiency. This move toward personalized health plans reflects a broader shift to patient-centred care, where risk assessments are specific, dynamic, and equitable (Dr. Josh Walters, 2024). Moreover, fairness-aware algorithms and explainability tools like SHAP values ensure transparency and help mitigate bias in pricing decisions (Kasaraneni, 2021). Thus, machine learning not only enhances premium accuracy but also supports ethical and sustainable insurance frameworks in diverse healthcare systems.

The rising cost of healthcare has emerged as a significant concern for both individuals and health insurance providers, especially in developing economies like Zimbabwe. For medical aid societies such as CIMAS, setting fair, accurate, and sustainable insurance premiums has become increasingly complex due to the dynamic nature of healthcare needs, economic instability, and diverse member health profiles. Traditional actuarial approaches, which primarily rely on demographic variables such as location, gender, and age, are proving lacking in adequately capturing the nuanced interplay between individual health conditions and utilization patterns(Kasaraneni, 2021). Consequently, there is a growing need for more sophisticated, data-driven approaches to predict healthcare costs and personalize premiums accordingly.

Machine learning (ML) has emerged as a transformative tool in addressing these challenges by enabling predictive analytics based on historical data. ML algorithms such as Random Forest, XGBoost, and deep learning architectures have demonstrated superior performance in modeling non-linear relationships and detecting hidden patterns in healthcare data (Kapse *et al.*, 2025; Patil *et al.*, 2024). These algorithms allow insurers to move beyond static demographic variables and instead leverage electronic health records (EHRs), claims history, lifestyle indicators, and chronic disease profiles to forecast individual healthcare costs with greater accuracy.

Recent studies confirm the effectiveness of ML in enhancing premium prediction and risk assessment accuracy. For instance, gradient boosting models have shown significant improvements in mean squared error and R-squared values when compared to linear regression models, making them highly suitable for estimating complex healthcare expenditures (Kapse *et al.*, 2025). The inclusion of explainable AI techniques, such as SHAP values, also contributes to increased transparency in model outputs, making it easier for stakeholders to understand how specific features impact premium predictions (Kapse *et al.*, 2025).

Moreover, machine learning can address persistent issues associated with traditional risk models, such as selection bias and data sparsity. Traditional models often fail to predict costs accurately for healthier individuals due to limited claims data, or underestimate expenses for individuals with multiple chronic conditions due to oversimplified linear assumptions (Kasaraneni, 2021). ML models, on the other hand, excel at modeling these non-linear and synergistic effects, offering more equitable and personalized pricing mechanisms.

Over the decades, health insurance has evolved significantly. Initially focused on covering hospital expenses, it has expanded to include preventive care, outpatient services, prescription drugs, and mental health services. The evolution reflects the changing healthcare landscape and the increasing complexity of medical needs. Today, the health insurance market includes a variety of plans, such as managed care, Health Maintenance Organizations (HMOs), and Preferred Provider Organizations (PPOs), each offering different levels of coverage and cost-sharing (Moreno *et al.*, 2020)

Premiums are central to the functioning of health insurance as they represent the monthly fee that policyholders must pay to maintain their coverage (Douven *et al.*, 2020). Premium pricing is a complex process influenced by several factors, including the insured's age, health status, geographic location, and the type of health plan selected. For instance, younger individuals generally pay lower premiums compared to older adults due to their generally lower health risks. Additionally, insurers consider historical claims data, anticipated healthcare costs, and the overall risk profile of the insured population when determining premiums (Lamiraud and Stadelmann, 2020).

The role of health risk in premium pricing is particularly pronounced in the United States. A significant portion of the population has expressed dissatisfaction with premium costs, often citing fairness and affordability as primary concerns. For example, according to a 2020 Kaiser Family Foundation survey, 47% of respondents reported that they found their premiums to be

unaffordable. Insurers like Anthem and UnitedHealth Group have faced backlash from members who argue that their premiums do not accurately reflect their health risks or the value they receive from their plans. Common complaints include significant premium increases in response to individual health claims, leading many to feel that they are being penalized for their health conditions (Holahan *et al.*, 2023).

In response to these challenges, medical aid providers have employed various strategies to mitigate member dissatisfaction regarding premium pricing. Many insurers have adopted community rating systems, which establish premiums based on the overall health of the insured group rather than individual health risks (Wengle *et al.*, 2022). Holahan *et al.*, (2023) argue that this approach helps to prevent discrimination against those with pre-existing conditions, a practice that was particularly scrutinized under the Affordable Care Act (ACA). Additionally, some medical aid providers have introduced wellness programs that incentivize members to engage in healthier lifestyles, potentially reducing overall healthcare costs and, in turn, stabilizing premium rates (Plott,2020).

Examining the context of the United Kingdom and China reveals similar challenges in health risk assessment and premium pricing. In the UK, the National Health Service (NHS) provides comprehensive coverage to residents, which alleviates many issues related to premium pricing for public health services. However, private insurers such as Bupa and Aviva still face challenges with balancing risk assessment and affordability (Detten,2023). Keegan (2020) posits that in China, the rapid expansion of private health insurance has led to increased scrutiny of how premiums are set. With rising healthcare costs, insurers like Ping An and China Life carefully consider health risks in their pricing models to maintain competitiveness while ensuring access to care.

In South Africa, the medical aid industry also faces unique challenges related to premium pricing. Many members have raised concerns about the fairness of premium rates, especially as the cost of living rises. A report by the World Health Organisation (2022) indicated that nearly 60% of South African medical aid members felt that their premiums were too high relative to the benefits received. Providers such as Discovery Health and Momentum Health have responded by offering tiered plans, allowing members to select coverage levels that align with their financial situations and health needs. Despite these efforts, complaints persist, with many members feeling that their premiums do not reflect their health status or usage of services (Adebisi *et al.*, 2023).

The situation in Zimbabwe's medical aid industry highlights similar struggles, particularly for providers like CIMAS. Mhazo (2023) indicates that CIMAS members have expressed significant concerns over premium increases, with contributions reportedly jumping from \$148 in May 2023 to \$304 in June 2023 an increase of over 100%. Members have voiced frustrations, claiming that CIMAS was short-changing them, particularly since their premiums are pegged in US dollars. Bhekimpilo Ntini, CIMAS's group marketing officer, acknowledged that the premium hikes primarily affected members with adverse claim ratios, illustrating the tension between risk assessment and member satisfaction.

The ethical and practical implications of charging uniform premiums for clients with varying health risks cannot be overlooked. Clients often argue that it is unfair to impose the same premiums on individuals with differing health profiles and needs (Zinyemba and Chabikisa, 2021). This sentiment underscores the necessity for Machine Learning-based predictive model that establishing premiums in line with the expected healthcare utilization costs for an individual or group on individual health data and associated cost. Such a system would not only enhance fairness in premium pricing but also improve member satisfaction by aligning costs with actual health risks.

The importance of premium pricing also extends beyond individual satisfaction as it directly impacts the performance of service providers. Striking a balance between fair premium pricing and the financial sustainability of health insurance providers is therefore essential. Insurers must navigate the delicate interplay between risk assessment, member satisfaction, and operational viability.

Therefore, the purpose of this study is to design a machine learning predictive model for personalized premium forecasts that can be mapped into a tiered pricing system, aimed at addressing the challenges in premium pricing while enhancing equity and client satisfaction within the health insurance landscape. Through leveraging advanced analytics and data-driven decision-making, this study seeks to contribute to a more equitable and efficient health insurance model, particularly in the context of CIMAS and the broader medical aid industry in Zimbabwe.

1.2 Problem statement

Health insurance providers continuously struggle to effectively evaluate the health risks of prospective clients. Conventional pricing models, which depend mainly on demographic details like age, gender, and location, are increasingly seen as inadequate because they fail to

account for the complex relationships between personal health factors and how individuals use healthcare services. These traditional models are also prone to limitations such as insufficient individual health data, which can result in flawed risk evaluations. In contrast, machine learning provides a powerful alternative by processing large-scale datasets and uncovering intricate patterns that conventional methods may miss (Kasaraneni, 2022).

Despite the crucial role of health insurance in providing financial protection for healthcare costs, many members experience dissatisfaction due to premium rates that do not adequately reflect individual health risks, leading to perceptions of unfairness and affordability issues. CIMUS has been grappling with the sustainability of its medical aid fund, as recent trends show claim ratios exceeding 105% and utilization rates approaching 40%. The organization mandates that at least 80% of contributions be allocated to claims and quality improvement, leaving only 20% for administrative expenses. However, this balance is being disrupted by members who utilize resources disproportionately compared to their contributions, leading to significant financial strain on the fund (Sunday Mail, 2023). CIMAS members exacerbate these concerns, prompting complaints that the pricing model inadequately considers their unique health profiles. Consequently, there is a pressing need for a Machine learning predictive model that predicts health utilisation cost and facilitate personalized insurance premiums, ensure fairness, enhance member satisfaction, and improve the overall performance of medical aid providers.

1.3 Conceptual framework

The conceptual framework for this study is built upon a systematic representation of the key variables involved in designing a machine learning predictive model for personalized insurance premiums.

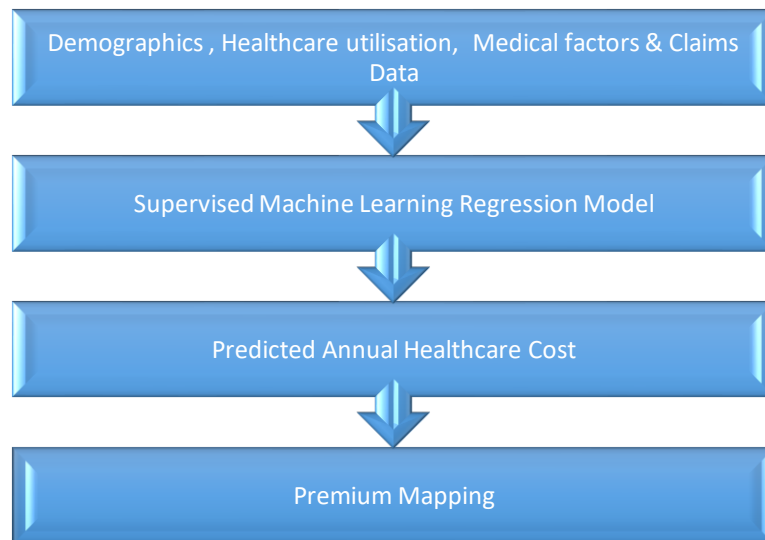


FIGURE 1 CONCEPTUAL FRAMEWORK

1.4 Research objectives

1. To design a machine learning predictive model for personalized insurance premiums.
2. To investigate the relationship between individual health factors, healthcare utilization patterns, and healthcare costs, on determining personalized premium pricing in health insurance.
3. To evaluate and compare the effectiveness of advanced machine learning algorithms in predicting personalized medical insurance premiums.

1.5 Research questions

1. What machine learning algorithms can be effectively employed to design a model for determining premium pricing for medical aid plans at Cimas?
2. What is the relationship between individual health factors, healthcare utilization patterns, and healthcare costs in determining personalized premium pricing in health insurance?
3. What is the comparative effectiveness of advanced machine learning algorithms in predicting personalized medical insurance premiums?

1.6 Significance of the study

The designing of a Machine Learning predictive model for personalized insurance premiums holds significant implications for various stakeholders within the healthcare and insurance sectors.

1.6.1 Insurance Providers

The primary beneficiaries of this study are insurance providers, such as CIMAS and similar organizations. By implementing a machine learning system, these providers can enhance the accuracy of risk assessments and optimize their premium pricing processes. This leads to more equitable premium structures that reflect individual health risks, thereby improving financial sustainability. The ability to adjust premiums in real time based on ongoing health assessments allows insurers to respond dynamically to changing health profiles, ultimately reducing the risk of financial loss due to inaccurate pricing.

1.6.2 Policyholders

The study significantly benefits members of health insurance plans by fostering a more personalized approach to premium pricing. When premiums are tailored to reflect individual health risks, members are likely to perceive their insurance coverage as fairer and more valuable. This can enhance overall member satisfaction and engagement, as individuals feel that they are being charged by their actual health status. Additionally, by promoting healthier lifestyles through personalized feedback from health risk assessments, members may experience improved health outcomes.

1.6.3 Healthcare Systems

This study contributes to broader healthcare systems by promoting preventive care and early intervention strategies. The integration of real-time health assessments encourages individuals to monitor their health more closely, potentially resulting in improved health management and lower healthcare costs over the long term, by aligning insurance incentives with health outcomes, the study supports the overarching goal of improving public health while simultaneously managing costs.

1.6.4 Academic and Research Communities

Ultimately, this study significantly enhances the academic and research communities by adding to the expanding literature on the applications of artificial intelligence (AI) in healthcare and insurance. It paves the way for future inquiries into the ethical considerations surrounding AI in health risk assessments. Additionally, it prompts further exploration into the long-term effects of personalized insurance pricing on both health outcomes and the dynamics of insurance markets. By examining these critical areas, the research not only informs current practices but also encourages a more nuanced understanding of how AI can transform healthcare and insurance while addressing ethical concerns. This comprehensive approach

invites ongoing dialogue and investigation, ensuring that the advancements in AI are aligned with the best interests of individuals and society as a whole.

1.7 Delimitations

This study was framed by specific delimitations that defined its scope and boundaries. Firstly, the research focused exclusively on the context of CIMAS medical aid in Zimbabwe. The study prioritized understanding the unique health profiles and risk factors relevant to the Zimbabwean population, thereby excluding broader demographic variables from other countries that may have influenced health risk assessments. Additionally, the study concentrated on adult members of health insurance plans, omitting considerations for children and adolescents, whose health risk factors and insurance needs differed significantly.

Secondly, the research specifically investigated the implementation of machine learning algorithms for cost prediction and premium calculation, without delving into the operational challenges that arose during the integration of such technology into existing insurance frameworks. While the study aimed to propose a model for personalised premiums, it did not cover the ethical implications surrounding data privacy and discrimination. By establishing these delimitations, the study maintained a focused approach, enabling a thorough examination of the variables directly related to the designing and application of the Machine Learning predictive model within the specified context.

1.8 Chapter summary

In this first chapter, the study provided a comprehensive overview that began with the background, setting the stage for the necessity of personalized premium pricing in the health insurance landscape. Following this, the problem statement outlined the challenges inherent in current pricing models. The chapter further presented the conceptual framework that illustrated the relationships among key variables, along with the objectives and research questions that guided the investigation. Additionally, it discussed the significance of the study, highlighting its potential benefits for various stakeholders, and concluded by delineating the delimitations that defined the study's scope and boundaries.

CHAPTER 2: LITERATURE REVIEW

2.0 Introduction

The next sections will critically examine current literature specifically related to the objectives of this study, in terms of the impacts of these AI technologies on data accuracy, risk profiling, individualized medical plans, and predictive analytics. The review will be ended with empirical literature to provide insights into real-world uses and experiences of machine learning in cost prediction, including successes and pitfalls encountered in various healthcare environments.

2.1 Theoretical Framework

The theoretical framework for this study is anchored on three foundational theories, Machine Learning Theory, Expected Utility Theory and Actuarial Fairness Theory. Machine Learning Theory underpins the application of advanced algorithms to analyse complex datasets, enhancing the accuracy of health risk assessments and enabling personalized insurance pricing. Expected Utility Theory provides a lens through which to understand consumer decision-making under uncertainty, illustrating how individuals evaluate insurance options based on the expected benefits and risks associated with their health profiles. Together, these theories create a comprehensive framework for analysing the impact of Machine Learning technologies on cost prediction and insurance premium pricing, guiding the exploration of the key objectives of this research. This framework specifically supports the development of a supervised machine learning model that predicts annual healthcare costs, ultimately aiming to improve the personalization of premiums for medical aid plans.

2.1.0 Machine Learning Theory

Machine Learning Theory includes a diverse array of algorithms and statistical models that empower computers to learn from data and make predictions (Mitchell, 1997). This theoretical framework is particularly relevant in healthcare, where the ability to analyse large and complex datasets has become increasingly important. The transformative potential of machine learning in health insurance lies in its capacity to derive actionable insights from data, thereby enhancing the accuracy and efficacy of health risk assessments. As insurers increasingly adopt machine learning techniques, understanding this theory becomes essential for evaluating their impact on risk profiling and premium pricing. (Bishop, 2006; Zhang *et al.*, 2022).

One of the primary applications of machine learning in health insurance is its role in risk profiling. By utilizing sophisticated algorithms, insurers can create detailed risk profiles for policyholders that consider a multitude of variables, including medical history, lifestyle choices, and demographic factors. Research by Chiu *et al.*, (2021) demonstrates that machine learning models can achieve higher accuracy in predicting health cost compared to traditional

actuarial methods. A study by Zhang *et al.* (2020) also highlights how machine learning enhances personalized insurance products by accurately assessing risk, allowing insurers to set premiums that reflect the true likelihood of claims, thereby enhancing the fairness of the insurance model.

Deep learning, a subset of machine learning, has shown remarkable potential for analysing unstructured data, such as medical images and clinical notes. These advanced algorithms can identify patterns and correlations that may not be evident through conventional analysis, leading to improved risk assessments. Studies by Xiong *et al.* (2023) and Lee *et al.* (2022) highlight the effectiveness of deep learning models in detecting risk factors for chronic diseases, enabling insurers to implement proactive measures in their coverage plans. This capability not only improves the accuracy of risk assessments but also enhances the overall quality of care provided to policyholders, ultimately benefiting both the insurer and the insured.

However, the implementation of machine learning in health insurance also raises several challenges that must be addressed. Issues related to data quality, interpretability of algorithms, and the potential for bias in decision-making are critical considerations. A systematic review by Usman *et al.* (2023) underscores the importance of ensuring that machine learning models are trained on diverse datasets to avoid perpetuating existing disparities in healthcare access and outcomes. Insurers must prioritize fairness and transparency in their machine learning practices to build trust among consumers and avoid the pitfalls associated with biased algorithms.

Machine Learning Theory provides a robust framework for understanding the applications and implications of AI in health risk assessments and insurance premium pricing. By leveraging advanced algorithms, insurers can enhance their predictive capabilities, leading to more accurate and personalized insurance products. As the field continues to evolve, ongoing research should focus on maximizing the benefits of machine learning while addressing the ethical and practical challenges that arise in its application. This balance will be essential for fostering an insurance landscape that is both innovative and equitable.

2.1.1 Expected Utility Theory

Expected Utility Theory offers a compelling framework for understanding how individuals make decisions under uncertainty, particularly in the context of health insurance. According to this theory, individuals evaluate potential choices based on the expected utility of outcomes, considering both the probabilities of various scenarios and the associated benefits. This

decision-making process is particularly relevant in health insurance, where consumers must navigate complex choices regarding coverage options, premiums, and health risks. By understanding how policyholders assess their options, insurers can better tailor their products to meet consumer needs.

One key application of Expected Utility Theory in health insurance is its relevance to the impact of accurate data on consumer decision-making. When reliable and comprehensive information about an individual's health status is available, it enhances the expected utility perceived by policyholders. Research conducted by Svendsen *et al.* (2020) indicates that accurate risk assessments can lead to better-informed choices, as consumers can more effectively weigh the benefits of different insurance options against their health risks. This alignment between data accuracy and expected utility underscores the importance of integrating reliable data systems within health insurance frameworks.

The contribution of machine learning to predictive analytics further enhances the decision-making process under Expected Utility Theory. By analysing historical data, machine learning models can predict future healthcare costs and outcomes, allowing consumers to evaluate their insurance options more effectively. A study by Abdulazeem *et al.* (2023) illustrates how machine learning enhances predictive analytics, empowering policyholders to make decisions that align with their risk profiles. This capability reinforces the concept of expected utility, as individuals can weigh their choices based on informed predictions, leading to more satisfactory insurance outcomes.

Expected Utility Theory thus provides a comprehensive lens through which to analyse consumer behaviour in health insurance, particularly in relation to machine learning. The impact of data accuracy, the effectiveness of deep learning in risk profiling, the role of decision support systems in customization, and the contributions of machine learning to predictive analytics collectively enhance the expected utility for policyholders. Future research should continue to explore how these theories intersect, focusing on optimizing consumer decision-making in the evolving landscape of health insurance. By addressing the complexities and uncertainties associated with health risks, insurers can foster a more informed and equitable marketplace.

2.2 Conceptual Framework

The conceptual framework for developing a supervised machine learning model for personalized premium determination at CIMAS is built on three critical categories of input

features: demographics, medical data, and claims data. Each of these categories plays a crucial role in enhancing the accuracy of healthcare cost predictions and ensuring equitable premium pricing.

Demographic data, including age, gender, and socio-economic status, significantly influences health risks and healthcare costs. Research has shown that these factors are instrumental in shaping health outcomes. For example, a study by Kim *et al.* (2021) highlights how age is a strong predictor of healthcare utilization, with older adults typically incurring higher costs due to chronic conditions. Additionally, socio-economic factors can influence access to healthcare services and overall health status, as noted by O'Neill *et al.* (2020), who found that lower socio-economic status correlates with increased health risks and higher healthcare expenses. By incorporating demographic data into the machine learning model, insurers can better understand the risk profiles of policyholders and tailor premiums accordingly.

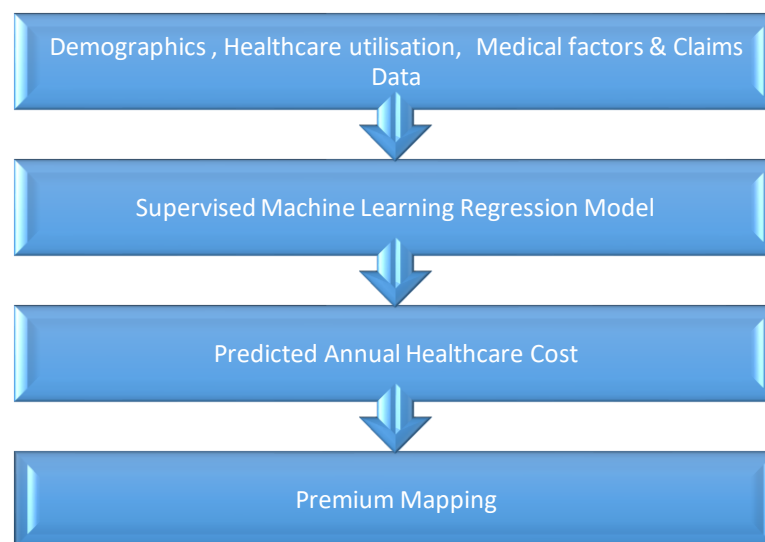


FIGURE 2 CONCEPTUAL FRAMEWORK 2

Medical data encompasses individual health histories and chronic conditions, which are critical for accurate risk assessment. Recent studies emphasize the importance of comprehensive medical information in predictive modelling. For instance, Nguyen *et al.* (2022) demonstrate that detailed medical histories, including past diagnoses and treatment patterns, improve the accuracy of risk predictions. Machine learning algorithms excel at analysing complex medical data, as highlighted by Zhang *et al.* (2021), who found that models trained on varied health indicators can significantly enhance predictions of future healthcare costs. This advanced analysis allows insurers to craft personalized insurance products that cater to the unique health needs of individuals, supporting the principle of actuarial fairness.

Claims data provides historical insights into healthcare utilisation and expenditures, serving as a vital resource for identifying trends that inform future cost predictions. Research indicates that analysing claims data can enhance the understanding of healthcare patterns and risks. A systematic review by Usman *et al.* (2023) emphasises the value of claims data in developing predictive models that account for previous utilization patterns, leading to more accurate forecasting of future costs. By leveraging this data, insurers can minimize the risk of adverse selection and ensure that premiums reflect the actual risks associated with policyholders.

The integration of demographic, medical, and claims data into a supervised machine learning model allows insurers to generate precise predictions of annual healthcare costs. This process is crucial for translating predicted costs into personalized premium amounts, fostering fairness in insurance pricing. A study by Habbal *et al.* (2024) reinforces this notion, showing that accurate cost predictions enable insurers to align premiums with individual health profiles, promoting equity within the insurance framework. Additionally, the application of machine learning techniques, such as regression models, enables the identification of complex relationships among input features, further enhancing predictive accuracy. Research by Lee and Kim (2022) demonstrates that advanced algorithms can uncover nuanced patterns in healthcare data, leading to improved risk assessments and more tailored insurance products.

2.3 Literature Relating to Objectives

2.3.1 To design a machine learning predictive model for personalized insurance premiums.

The development of predictive health risk scoring models using Supervised learning techniques have garnered considerable attention in recent years, especially as healthcare systems aim to utilize data to enhance patient outcomes. A study by Kourou *et al.*, (2021) illustrates the effectiveness of supervised learning algorithms in predicting health risks by utilizing historical patient data. The authors emphasize that by training models on labelled datasets, healthcare providers can identify patients at high risk for various conditions, such as diabetes and cardiovascular diseases. This predictive capability allows for timely interventions, which can significantly reduce morbidity and mortality rates. The findings align with those of Esteva *et al.* (2020), who also highlight the importance of using historical data to inform predictive models, thereby underscoring the potential of supervised learning in enhancing clinical decision-making.

Moreover, the integration of various supervised learning techniques, such as logistic regression, decision trees, and support vector machines, has been shown to improve the

accuracy of risk predictions. For instance, a comparative study by Reddy *et al.* (2022) found that ensemble methods, which combine multiple algorithms, outperformed individual models in predicting health risks among patients with chronic illnesses. This finding is particularly relevant for the Cimas Medical Aid context, where a diverse member population may exhibit varying health profiles. The study suggests that employing a combination of supervised learning techniques can lead to more robust and reliable risk scoring models, ultimately enhancing the personalization of healthcare interventions.

In addition to algorithm selection, the quality of input data plays a crucial role in the effectiveness of predictive models. Research by Alzubaidi *et al.* (2021) emphasizes the importance of data preprocessing and feature selection in developing accurate risk scoring models. The authors argue that incorporating relevant features, such as demographic information and previous health records, can significantly improve model performance. This perspective is echoed by Kourou *et al.* (2021), who highlight the necessity of using comprehensive datasets to train algorithms effectively. The emphasis on data quality underscores the need for healthcare organizations to invest in robust data management practices to ensure that predictive models yield accurate and actionable insights.

Furthermore, the ethical implications of predictive modelling in healthcare cannot be overlooked. Char *et al.* (2020) raise concerns about the potential for bias in AI-driven risk assessments, particularly when models are trained on datasets that do not adequately represent diverse populations. This issue is critical for Cimas Medical Aid, as the organization must ensure that its predictive models do not inadvertently perpetuate health disparities. Addressing these ethical considerations is essential for fostering trust in AI technologies and ensuring equitable access to healthcare services. The integration of fairness metrics into predictive modelling, as suggested by Reddy *et al.* (2022), could help mitigate these risks and promote more equitable health outcomes.

The application of predictive health risk scoring models also extends to resource allocation within healthcare systems. A study by Esteva *et al.* (2020) highlights how accurate risk predictions can inform healthcare providers where to allocate resources most effectively. By pinpointing high-risk patients, healthcare organizations can prioritize interventions and distribute resources more efficiently, ultimately resulting in better health outcomes and reduced costs. This strategic approach aligns with the goals of Cimas Medical Aid, which seeks to optimize its healthcare offerings for its members.

Moreover, the continuous evolution of machine learning algorithms presents opportunities for enhancing predictive health risk scoring models. Recent advancements in AI, such as the development of more sophisticated neural networks, have the potential to improve the accuracy and reliability of risk predictions. Research by Alzubaidi *et al.* (2021) suggests that incorporating deep learning techniques into predictive modelling could further enhance the identification of complex patterns in health data. This integration of advanced algorithms may lead to more nuanced risk assessments, allowing healthcare providers to tailor interventions more precisely to individual patient needs.

2.3.2 To investigate the relationship between individual health factors, healthcare utilization patterns, and healthcare costs, on determining personalized premium pricing in health insurance.

The identification of risk factors within the medical aid member population using supervised learning algorithms has emerged as a vital area of research in healthcare analytics. Recent studies have demonstrated the effectiveness of predictive modelling techniques in uncovering significant patterns in health data that are not readily apparent through traditional analysis. For instance, Alzubaidi *et al.* (2021) utilized decision trees to analyse health records of patients, revealing distinct subgroups with unique risk profiles. This approach allows healthcare providers to identify key risk factors that may contribute to adverse health outcomes, thereby enabling more targeted interventions. The findings underscore the potential of supervised learning to enhance the understanding of complex health behaviours within diverse populations.

Additionally, the application of logistic regression has been shown to provide valuable insights into the relationships between different health conditions. A study by Reddy *et al.* (2022) employed logistic regression to analyse the health data of a large cohort, identifying significant predictors of health risks among patients. The authors argue that this method not only reveals critical risk factors but also facilitates the exploration of comorbidities that may not be immediately evident. This perspective aligns with the goals of Cimas Medical Aid, as understanding the interplay between various health conditions can inform the development of comprehensive care plans that address the multifaceted needs of its members.

Moreover, the integration of supervised learning techniques with demographic data can enhance the identification of risk factors across different population segments. Research conducted by Kourou *et al.* (2021) highlights the importance of incorporating demographic

variables, such as age, gender, and socioeconomic status, into predictive analyses. By doing so, healthcare organizations can uncover disparities in health risks that may be influenced by social determinants of health. This approach is particularly relevant for Cimas Medical Aid, as it seeks to ensure equitable access to healthcare services for all members, regardless of their background.

The ethical implications of using supervised learning to identify risk factors also warrant consideration. Char *et al.* (2020) emphasize the need for transparency in AI-driven analyses, particularly when uncovering sensitive health information. The potential for misinterpretation of data or the reinforcement of existing biases poses significant challenges for healthcare organizations. To address these concerns, it is essential for Cimas Medical Aid to implement ethical guidelines that govern the use of supervised learning techniques, ensuring that the insights gained are used responsibly and equitably.

Furthermore, the identification of risk factors can inform preventive healthcare strategies, ultimately leading to improved health outcomes. A study by Esteva *et al.* (2020) highlights how uncovering significant risk factors can enable healthcare providers to implement targeted interventions that address specific health needs. For example, if predictive analyses reveal a subgroup of patients with high rates of hypertension, healthcare organizations can develop tailored programs to manage and mitigate this risk. This proactive approach aligns with the objectives of Cimas Medical Aid, which aims to enhance the health and well-being of its members through personalized care solutions.

In addition to traditional supervised learning methods, the use of ensemble techniques, such as random forests, has been shown to enhance the identification of risk factors. Research by Alzubaidi *et al.* (2021) demonstrates that ensemble methods can improve prediction accuracy by combining multiple models, allowing for a more comprehensive understanding of underlying patterns in health data. This technique can facilitate the exploration of relationships between various health indicators, providing valuable insights for healthcare providers seeking to understand the multifactorial nature of health risks.

Moreover, the continuous advancement of supervised learning algorithms presents opportunities for further enhancing the identification of risk factors. Recent developments in machine learning, such as gradient boosting machines, have shown promise in uncovering complex patterns in high-dimensional data. A study by Reddy *et al.* (2022) suggests that incorporating these advanced techniques into health analytics could lead to more nuanced

insights into patient populations. This integration of cutting-edge algorithms may enable Cimas Medical Aid to refine its understanding of risk factors, ultimately leading to more effective healthcare interventions.

2.4 Empirical Literature

This section offers an overview of empirical work that has been conducted with regard to the application of machine learning in cost prediction and insurance premium personalization.

There have been significant advances in North America in terms of machine learning health risk assessment systems as made apparent in recent studies. For example, a study by Habbal et al. (2024) indicates the application of machine learning methods in analysing electronic health records (EHRs) for improving health risk predictions. As per their findings, these pairings of advanced analytics are able to predict risks better, which in turn enables the insurers to tailor premiums based on individualized health profiles. But the report also identifies data interoperability and privacy issues, suggesting that these are addressed in order to ensure that the potential of Machine learning is fully realized in this area.

On the other hand, a McKinsey & Company (2022) analysis presents the limitations for organizations looking to implement machine learning solutions. While recognizing the potential benefit, the analysis points out that the majority of healthcare providers are hindered by data integration and quality, which impede effective AI technology usage. Such disparity in findings underscores the need for further work in the field of best practice data integration and management to enhance the effectiveness of AI-enabled health risk assessments.

European academics have also contributed to discourse regarding AI in health risk assessment. Beck *et al.* (2022) examined the ethical dimensions of AI within clinical decision-making and insurance pricing. To them, although AI can improve accuracy, it also raises concerns over transparency and accountability, particularly concerning algorithm development and validation. Such an outlook is needed to understand the broader implications of AI adoption in healthcare and insurance sectors. Again, however, no practical solutions are noted in the research towards resolving these ethical questions, i.e., no gap completed with suggestions for practitioners to implement.

Kauffman et al. (2023) further released a broad overview of AI application in health insurance, this time stressing the built-in biases within algorithm-based pricing models. Their work indicates that if biased historical data are used to train AI algorithms, they will reinforce existing health insurance premium imbalances. Testing and validation protocols are mentioned

as being necessary, but the authors fail to make significant strides in outlining how insurers can actively strip their models of bias. This lack of real-world direction is an opportunity waiting to be seized by further research focused on establishing frameworks for the ethical use of AI in insurance.

In Asia, scholars have explored the nexus of health risk assessment and AI to develop personalized insurance. Lee and Kim (2022), for instance, discuss the use of AI algorithms to predict health risks and adjust insurance premiums according to them. Their research indicates that AI is capable of enhancing risk estimation accuracy significantly but brings in data bias and privacy-related ethical concerns. The study draws the attention of insurers to transparency in how AI algorithms operate and to the necessity of training data sets. However, it does not provide definite instructions on how insurers can be ethically compliant, and an aspect to be further studied is left to be filled in.

Also, Chen *et al.* (2021) discusses the effect of the integration of AI into health insurance programs in terms of emphasizing the benefit of improved risk stratification and individualized premium charging. While hopeful, the study also cites the lack of regulation schemes that could ensure the secure and effective utilization of AI across the sector. This gap in regulatory oversight opportunities lie ahead for future research to propose frameworks that can ensure that the application of AI is ethical in health insurance and does not threaten to erode consumer rights or privacy.

Research in Africa such as work by Moyo *et al.* (2021) highlights how the affordability and accessibility of health insurance products are to be facilitated with AI technologies. Their research shows that AI-driven health risk evaluations can provide valuable insights pertinent to regional health matters, thus allowing insurers to create more competitive and relevant premium products. But the research also identifies significant adoption hurdles such as poor technology infrastructure and data governance issues in much of the continent. These findings project a critical need for research focused on surmounting them in a bid to facilitate the effective utilization of AI in health risk assessment within the African context.

Moreover, another study by Chikozho *et al.* (2022) focuses on the opinions of healthcare experts on the utilisation of AI in individualizing insurance premiums. The findings indicate widespread optimism in AI potential but also reveal data accuracy and privacy concerns with AI-driven assessments. Both perceptions call for involving stakeholders in discussions about AI use to allay concerns and enable proper adoption. However, the research lacks quantitative

data to affirm the qualitative findings, leaving a window of opportunity for future studies to explore the effect of AI solutions in real contexts across Africa.

2.5 Gap Analysis

The literature indicates some of the most remarkable gaps that illustrate areas missed by previous researchers in the adoption of Machine Learning cost prediction and customised insurance premiums. Though most of the research revolves around the technical capability of machine learning and how it will increase precision, a tremendous lack of attention is given to the ethical implications of these technologies, particularly bias and transparency in algorithmic decision-making. Research also fails to fully address the infrastructural challenges faced in low-resource settings, such as in Africa, where technology limitations will hinder the utilisation of AI. Moreover, while there are reports that cite the importance of data quality, they do not provide actionable guidelines for ethical compliance or practical methods for data governance.

Despite the growing interest in applying machine learning (ML) for premium pricing in health insurance, significant research gaps remain, particularly in the context of developing countries such as Zimbabwe. Existing literature primarily focuses on developed markets with robust health data ecosystems and advanced digital infrastructure (Khan et al., 2020; Wang et al., 2021). While machine learning techniques such as Random Forests, Gradient Boosting, and Deep Neural Networks have demonstrated high predictive performance in risk assessment and cost prediction (Dinh et al., 2022; Shi et al., 2020), their practical adaptation to local contexts in Sub-Saharan Africa remains underexplored.

In Zimbabwe, and by extension CIMAS Medical Aid Society, lacks documented case studies that explore the deployment of supervised machine learning models for real-time premium personalization using health risk factors such as age, BMI, chronic conditions, and healthcare usage behaviour. This creates a clear gap in the literature, particularly in developing frameworks that align ML-based pricing models with local regulatory, operational, and ethical realities (Nyatsanza & Mungwari, 2024).

Therefore, this study aims to bridge these gaps by developing a machine learning model tailored to the Zimbabwean health insurance context that uses both structured healthcare usage and health factors data to predict healthcare costs and recommend personalized premiums, enhancing fairness, affordability, and predictive accuracy.

2.6 Chapter Summary

This chapter provided a comprehensive literature review to the study, beginning with the analysis of the theoretical foundation upon which the application of machine learning was applied in health cost estimation and medical aid planning. Following this, the conceptual structure was detailed to illustrate the relationship of various technological components such as electronic health records, deep learning and machine learning, and how they help enhance healthcare delivery. The following sections of the literature were critically appraised for direct linkage to the aims of the study, such as the effect of these AI technologies on the accuracy of the data, risk stratification, customized medical plans, and predictive modelling. Next, empirical literature was explored to obtain knowledge regarding real-world use and implications of AI in health risk assessment and successes and pitfalls faced in different health care environments.

CHAPTER 3: RESEARCH METHODOLOGY

3.0 INTRODUCTION

This chapter outlines the methodological framework adopted in designing a machine learning model aimed at generating personalized insurance premiums. It details the overall research design, data sources, population sampling strategies, tools for data collection, ethical considerations, and approaches to ensuring data validity and reliability. A central focus of this chapter is the systematic data preparation process, which includes comprehensive feature engineering strategies designed to boost the model's predictive capability. These processes involve data cleansing, normalization, and the construction of new variables that reflect complex interactions among individual health indicators. Following data preparation, the study applies various supervised learning algorithms and uses cross-validation techniques such as k-fold validation to mitigate overfitting and improve model robustness. Model performance is assessed using multiple evaluation metrics. These include Mean Squared Error (MSE), R-squared (R^2) for assessing predictive accuracy, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC), particularly relevant when distinguishing between risk-based expenditure categories (Zhang *et al.*, 2022; Reddy *et al.*, 2022). This comprehensive approach ensures that the developed model is both reliable and applicable to real-world health insurance contexts.

3.1 Research Methodology

Research methodology refers to the structured framework and strategies employed to carry out a study systematically. It encompasses the techniques and procedures used for gathering and analysing information in order to enhance knowledge or gain a deeper understanding of a specific issue (University of Newcastle Library Guides, 2019). The three core methodological approaches widely recognized in academic research are Quantitative, Qualitative, and Mixed Methods Research (Creswell & Creswell, 2018).

Mixed Methods Research integrates both qualitative and quantitative techniques within a single study. According to Dawadi, Shrestha, and Giri (2021), this approach involves distinct philosophical foundations and allows for data to be collected and analysed using various tools, offering a more comprehensive perspective.

Qualitative Research is primarily exploratory, aimed at providing rich, detailed insights into a phenomenon without necessarily involving numerical data. As described by Oranga and Matere (2023), it focuses on understanding human experiences, behaviours, and perceptions,

addressing questions such as “how” and “why” rather than “how much.” Quantitative Research, on the other hand, deals with the collection and interpretation of numerical data. This method is used to identify patterns, establish relationships, make predictions, and generalize findings across populations (Bhandari, 2020).

The research methodology employed in this study was quantitative in nature, which allowed for the systematic collection and analysis of numerical data. This approach was justified by the need to quantify health risks and evaluate the effectiveness of predictive models. The quantitative methodology facilitated the use of statistical techniques to draw objective conclusions based on the collected data from Cimas Medical Aid member database. As demonstrated by Murrile *et al.*, (2021), quantitative research is particularly effective in healthcare settings, where the identification of patterns and correlations can lead to significant advancements in patient management and intervention strategies. By focusing on measurable outcomes, the study aimed to provide concrete evidence for the effectiveness of machine learning predictive model on personalizing premiums based on the predicted healthcare cost.

3.2 Research Design

A research design provides a structured, step-by-step framework that guides researchers in systematically conducting a scientific investigation. It outlines the methods and strategies used to address a research problem effectively. The primary types of research designs include exploratory, descriptive, and explanatory (or causal) approaches (Saunders *et al.*, 2019). For the purposes of this study, the case study design was adopted, as the research focuses specifically on a single insurance company operating in Zimbabwe. This approach is appropriate when a detailed and contextual analysis of a particular organisation or phenomenon is needed. As noted by Kothari and Garg (2020), case study design supports comprehensive and in-depth investigation, enabling researchers to gain holistic insights into the subject matter.

3.3 Target Population

The target population for this study consisted of 10,000 members of Cimas Medical Aid according to the company’s marketing department. This population size represented a diverse demographic in terms of age, gender, and health status, which was essential for capturing a wide range of health profiles. The importance of diversity in health research is well-documented; studies like those conducted by Alzubaidi *et al.* (2021) highlight that a varied participant pool enhances the generalizability of findings. By focusing on such a substantial population, the study aimed to ensure that the insights gained would be relevant and applicable

to various subgroups within the Cimas membership, thus providing a comprehensive understanding of the health risks faced by different demographic segments.

3.3.1 Sampling Frame and Procedure

The sampling frame from which the sample was drawn included a complete list of Cimas Medical Aid members, maintained in an electronic database by the organization. This database contained essential demographic and health information for each member, enabling the researchers to select a representative sample effectively. Utilizing a well-defined sampling frame was crucial for the integrity of the research, as noted by Esteva *et al.*, (2020). The availability of detailed records allowed researchers to stratify the population based on relevant attributes, ensuring that the sample accurately reflected the characteristics of the target population. This comprehensive approach to sampling provided a solid foundation for the subsequent analyses and findings.

3.3.2 Sampling Procedure

The sampling procedure employed was stratified random sampling, which involved dividing the Cimas member population into distinct subgroups or strata based on relevant characteristics, such as age, gender, and health status. This method was justifiable as it ensured representation across different demographics, an approach highlighted by Reddy *et al.*, (2022). By randomly selecting participants from each stratum, the researchers aimed to minimize bias and enhance the generalizability of the findings. This stratified approach allowed for a more detailed analysis of health risks within specific subgroups, ultimately leading to more tailored healthcare interventions. The careful stratification of the sample was particularly beneficial for Cimas Medical Aid, as it aimed to develop insights that could address the unique needs of its diverse membership.

3.4 Data Sources

Data for this study were sourced exclusively from secondary sources. Secondary data were obtained from the Cimas Medical Aid database, which included historical health records, demographic information, and previous claims data. This data source has been

shown to provide valuable insights into health trends and risk factors, as noted by Alzubaidi *et al.*, (2021). The secondary data provided a rich context for understanding existing health issues within the member population. By leveraging these secondary sources, the study aimed to

enhance the robustness of its findings and provide more actionable recommendations for healthcare providers.

3.5 Research Instruments

The study utilized a systematic data extraction protocol to gather relevant information from the Cimas Medical Aid database, which included health records, demographic details, and claims data. Machine learning software, such as Python was employed to conduct comprehensive analyses and build predictive models based on the extracted data. Documentation and coding manuals ensured consistency and accuracy in variable definitions and coding throughout the analysis process. Additionally, a literature review framework provided context for the findings and enhanced their interpretation, effectively supporting the study's objectives.

3.6 Research Process Model

The research employed the Cross Industry Standard Process for Data Mining (CRISP-DM) model. The process model was used for this study because of its wide spread use in data analysis and mining, extensive backtracking and its flexibility. CRISP-DM is an industry-wide proven way or process model of guiding a data science or machine learning project, (Hotz, 2023). Furthermore, the researcher chose CRISP-DM because it is cost effective as it includes a number of processes that outlines simple machine learning tasks and also the tasks are well established across the industry. CRISP-DM also encourages the best practices and it allows the replication of projects. In addition, CRISP-DM provides a uniform framework for managing and planning a machine learning project. According to Rodrigues (2020), CRISP-DM is a 1996 invention that plans, organizes and executes data mining (machine learning) projects. Therefore, for the purposes of this research, the CRISP-DM constitutes the process model that outlines the phases of the supervised machine learning project, tasks connected to each phase and also the relationships between the tasks, whilst providing an overview of the data mining (machine learning) life cycle. According to Hotz (2023), CRISP-DM framework structures the data mining process into six distinct phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment, guiding the project from problem definition to practical implementation.

3.6.1 Business Understanding

CIMAS Medical Aid Society in Zimbabwe offers health insurance through tied pricing packages, where members choose from a fixed set of plans such as Secure Prime, Secure

Premium, Secure Private, Secure Essential, Secure Primary, and Health Save. Each package has predefined rates that vary by category corporate vs. individual and adult vs. child/student. For instance, the monthly individual adult rate ranges from \$253 under Secure Prime to as low as \$7 under Health Save, reflecting significant variation in access levels and service benefits.

The business objective of this study is to leverage machine learning (ML) to build a predictive model that estimates an individual's expected annual claim cost. Although CIMAS does not assign individualized premiums, such a model can inform personalized package recommendations. This ensures that members are matched with a plan that aligns with their predicted healthcare needs and financial capacity, ultimately improving fairness, affordability, and sustainability in the current pricing system.

The business hypothesis is that by training supervised ML models on historical claims data and policyholder attributes (e.g., age, gender, bmi, chronic conditions, smoking status, number of hospital visits, etc.), the model can predict individual risk profiles and healthcare utilization patterns. These insights can then guide CIMAS in classifying members into the most suitable package, rather than relying solely on general demographic grouping or member preference.

Moreover, this approach can help optimize tier boundaries, identify outliers (e.g., low-risk members in high-tier plans or high-risk members in low-tier plans), and inform future adjustments to package pricing. It also enables better cost forecasting, risk pooling, and preventive care interventions, especially for high-risk policyholders. In this way, the predictive model serves as a decision-support tool, allowing CIMAS to transition from static group-based assignment to a data-informed, risk-aligned classification within the existing tier-pricing framework, without disrupting regulatory or operational structures.

3.7 Research Ethics

In conducting the study, several research ethics were adhered to, ensuring the integrity and confidentiality of the data. First, all data were anonymized to protect the identities of Cimas Medical Aid members, thereby safeguarding their privacy. It was also mandated that the data be used solely for research purposes and not for any commercial gain. Additionally, the study followed principles of transparency by clearly documenting the data sources and methodologies employed in the analysis. Finally, ethical approval was obtained from CIMAS to ensure compliance with established standards for research involving health data.

3.8 Validity and Reliability

In the context of machine learning research, validity and reliability can be assessed using training error and test error, respectively. As explained by Mike and Hazzan (2022), training error measures the model's performance on the same dataset it was trained on, indicating how well the model captures the patterns in the training data therefore serving as a measure of validity. Conversely, test error evaluates how accurately the model performs on unseen data, which it has not encountered during training. This reflects the reliability of the model, as it demonstrates the model's ability to generalize to new inputs.

To ensure the validity and reliability of the research findings, several strategies were implemented regarding the secondary data used in the study. The data extraction process was conducted systematically to ensure that the information gathered from the Cimas Medical Aid database accurately reflected the relevant health profiles and risk factors. Various validation checks were employed to confirm the accuracy and completeness of the data, ensuring that discrepancies were minimized. Additionally, reliability was assessed through consistency checks across different data entries and historical records. These measures are consistent with the recommendations of Kourou *et al.*, (2021), who emphasized the importance of robust validation and reliability testing in health research to enhance the credibility of findings. By employing these strategies, the researchers aimed to ensure that the data would yield meaningful insights and contribute to the overall rigor of the study. Therefore, the researcher computed the training error and test error to confirm the reliability and validity of the research.

3.9 Data Analysis and Presentation

Python software was utilized for data analysis in this study due to its robust libraries and tools, which facilitated the processing and examination of the secondary data obtained from the Cimas Medical Aid database. Libraries such as Pandas and NumPy were employed for data manipulation and statistical analysis, while Scikit-learn was used for building predictive models. For presentation of the findings, visualization libraries like Matplotlib and Seaborn were leveraged to create clear and informative graphs and charts, effectively illustrating key trends and insights derived from the data analysis. This combination of analytical and visualization tools ensured that the results were both rigorously analysed and readily accessible for interpretation. The Jupyter notebook was used as the Integrated Development Environment (IDE), to run the code for designing the model. The researcher chose the Jupyter notebook because of its simplicity to use and also that it allows the researcher to compile all the aspects

of the machine learning project in one place thereby making it easier to show the entire process of the project to the audience of this research.

3.9.1 Python libraries used for the research

A number of libraries were used in python in all stages of designing the model and these are as below:

Python Library	Library's task
Pandas	This library was used for data manipulation, data analysis and data cleaning. This was used by the researcher because of its ability to simplify data modelling and data analysis operations without needing to write a lot of code.
NumPy	This was used for the purposes of the research because of its efficiency in numerical calculations that involve arrays and also because of its ability to perform simple regression tasks.
Scikit-learn	This library was used by the researcher because of its ability in fitting and making predictions. The model which forms the basis of the research is a predictive one hence the use of this library.
Seaborn	This Python library cannot be used to calculate regression parameters; however, the researcher chose to use in the research due to its ability to visualize statistical results of the model and regression lines. It was used for the generation of heatmaps.
Plotly	The researcher used this library because it allows the further improvement in the visualization experience by using customizable interactive tools.

Matplotlib	This library was used by the researcher because of its ability to allow the visualization of data in numerous ways that include, histograms, scatterplots, bar charts, boxplots and error charts.
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TABLE 1 PYTHON LIBRARIES

3.9.2 The Dataset

The dataset contains 1,378 records and 18 variables, including the target variable total claim cost. It encompasses insurance claim information, including patient demographics (age, sex, BMI), lifestyle factors (smoker, exercise), medical history (blood pressure, diabetes, comorbidities), healthcare utilization (number of visits, prescriptions), and the total claim cost. This data was used to analyse factors influencing insurance claim costs and potentially predict future expenses. The dataset was saved in a CSV file.

```
3]: print("There are {:,} observations and {} columns in the data set.".format(data.shape[0], data.shape[1]))
    print("There are {} missing values in the data.".format(data.isna().sum().sum()))

There are 10,000 observations and 19 columns in the data set.
There are 0 missing values in the data.
```

FIGURE 3 THE DATASET

Description of Variables

TABLE 2 DESCRIPTION OF VARIABLES

Variable	Description
Age	Age of the insured individual (in years)
BMI	Body Mass Index
Gender	Gender (0 = Male, 1 = Female)
Location	Coded location of the insured individual
Chronic Meds	Whether on chronic medication (0 = No, 1= Yes)
Smoking Status	Smoking status (0 = Non-smoker, 2 = Smoker)
Regular Exercise	Regular physical activity (0 = No, 1= Yes)
Diabetes	Diagnosis of diabetes (0 = No, 1 = Yes)
Blood Pressure	Hypertension condition (0 = No, 1 = Yes)
Heart Problem	History of cardiovascular conditions (1 = No, 1= Yes)
Asthma	Diagnosed with asthma (0 = No, 1= Yes)

Dependencies	Number of dependents covered under the plan
Plan Type	Type of medical aid plan selected (coded: e.g. 1 = health save, 6 = Prime)
Payer Type	Payment arrangement (0 = Individual, 1 = Employer, etc.)
Inpatient Visits	Number of inpatient hospital visits
Outpatient Visits	Number of outpatient visits
Revisits	Number of emergency room visits
Num Prescriptions	Number of prescribed medications
Num Prescriptions	Total annual healthcare claims cost (in USD)

Dataset Loading

The researcher loaded the dataset into the Jupyter Notebook using the below Python code:

```
data = pd.read_csv("C:\\Users\\user\\Desktop\\CIMAS2024_health_insurance_data.csv")
data.head()
```

Age	BMI	Gender	Location	Chronic_Meds	Smoking_Status	Regular_Exercise	Diabetes	Blood_Pressure	Heart_Problem	Asthma	Dependencies	Plan_Type	Payer_Type
54	37.2	Male	Bulawayo	1	0	1	0	0	0	0	2	Secure Essential	Adult: Individual
44	28.4	Male	Mutare	0	1	0	0	0	0	0	0	Health Save	Adult: Corporate
30	34.6	Female	Mutare	0	0	1	0	0	0	0	3	Secure Premium	Adult: Corporate
58	29.5	Male	Mutare	0	1	1	0	1	0	0	4	Secure Private	Adult: Individual
23	29.3	Female	Gweru	0	0	0	0	0	0	0	3	Secure Prime	Child: Individual

FIGURE 4 DATASET LOADING

3.8 Feature Engineering

According to Logunova (2022), feature engineering is the process of creating and transforming features that can be used in the training of a machine learning model. It involves sorting features, imputing missing values, transforming and discretizing numerical variables, dismissing duplicate records, removing or censoring outliers and modification of some data columns so as to obtain new features that are relevant to the study. If done properly, feature engineering can improve the efficiency of a model and also the accuracy of its results.

3.8.1 Data Clean-up

The researcher started the data clean-up process by checking for missing values and duplicate records. Missing values can misclassify a machine learning model therefore it is important to analyse missing data and correct them using the appropriate imputation method. Since the

dataset had no missing values as shown in figure 3 the researcher went on with exploratory data analysis.

3.9 Exploratory Data Analysis

According to Prajwal (2022), Exploratory Data Analysis (EDA) involves a process that provides a brief understanding of the collected data. EDA was applied in the research so as to summarize the key insights of the data and also to investigate the data. EDA was used by the researcher to check for patterns, outliers and trends in the collected data. EDA allowed the researcher to run descriptive statistics and also to visualize the data in graphs for easier understanding. Below is the EDA as performed by the researcher;

```
data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                   10000 non-null  int64
1   BMI                   10000 non-null  float64
2   Gender                10000 non-null  object
3   Location              10000 non-null  object
4   Chronic_Meds          10000 non-null  int64
5   Smoking_Status        10000 non-null  int64
6   Regular_Exercise      10000 non-null  int64
7   Diabetes              10000 non-null  int64
8   Blood_Pressure        10000 non-null  int64
9   Heart_Problem         10000 non-null  int64
10  Asthma                10000 non-null  int64
11  Dependencies          10000 non-null  int64
12  Plan_Type             10000 non-null  object
13  Payer_Type            10000 non-null  object
14  Inpatient_Visits      10000 non-null  int64
15  Outpatient_Visits     10000 non-null  int64
16  ER_Visits             10000 non-null  int64
17  Num_Prescriptions     10000 non-null  int64
18  Total_Claim_Cost      10000 non-null  float64
dtypes: float64(2), int64(13), object(4)
memory usage: 1.4+ MB
```

```
[4]: data.nunique()

[4]: Age                50
     BMI                241
     Gender              2
     Location            5
     Chronic_Meds        2
     Smoking_Status      2
     Regular_Exercise    2
     Diabetes            2
     Blood_Pressure      2
     Heart_Problem       2
     Asthma              2
     Dependencies        5
     Plan_Type           6
     Payer_Type          4
     Inpatient_Visits    10
     Outpatient_Visits   15
     ER_Visits           8
     Num_Prescriptions   26
     Total_Claim_Cost    9994
     dtype: int64
```

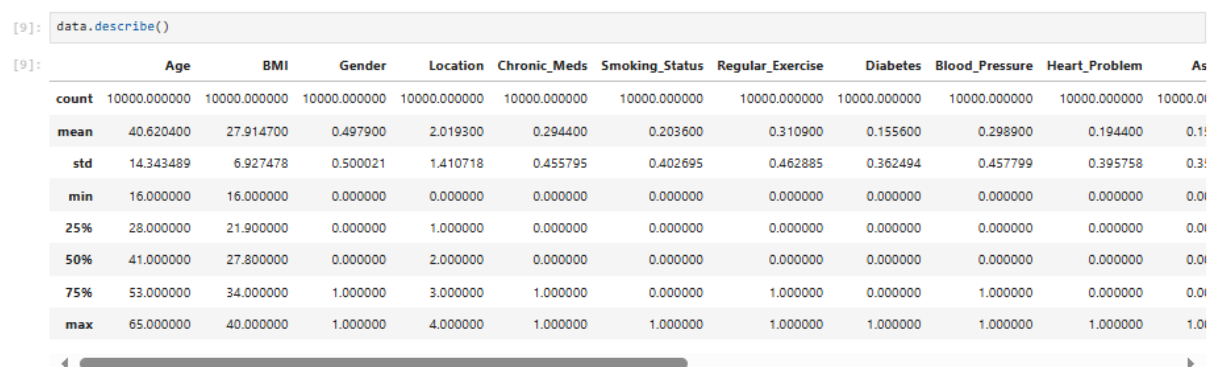
FIGURE 5 DATASET INFORMATION

From the above, it can be seen that the dataset had 10 000 entries, and the code above assists in obtaining further details about the data type for each variable in the dataset. From the result above, we can deduce that four of the variables were categorical (labelled as “object”) and the remaining were numerical (labelled as “int” and “float”). Additionally, all columns have non-null entries, indicating that the dataset is complete. The memory usage of 1.4 MB suggests that it is efficient for processing. This structure allows for various analyses, especially considering the need to encode categorical variables for machine learning applications.

3.10 Descriptive Statistics

Using the figure below, we can observe a detailed statistical summary of the dataset. Each variable is accompanied by key descriptive statistics, including count, mean, standard deviation (std), minimum (min), 25th percentile (25%), median (50%), 75th percentile (75%), and maximum (max) values. For instance, the mean age is approximately 40.2 years, with a standard deviation of 14.3, indicating some variability in age among the entries. The bmi variable shows a mean of around 27.91, suggesting that, on average, individuals in this dataset may be classified as normal weight.

Additionally, the total_claim_cost variable has a significantly higher range, with a maximum value of 62 930.56 which indicates a wide disparity in claims among the dataset entries. This summary provides valuable insights into the central tendencies and variability of the data, which was used for further analysis or modelling efforts, checking for multicollinearity and skewness in data distribution.



```
[9]: data.describe()
```

	Age	BMI	Gender	Location	Chronic_Meds	Smoking_Status	Regular_Exercise	Diabetes	Blood_Pressure	Heart_Problem	As
count	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000
mean	40.620400	27.914700	0.497900	2.019300	0.294400	0.203600	0.310900	0.155600	0.298900	0.194400	0.194400
std	14.343489	6.927478	0.500021	1.410718	0.455795	0.402695	0.462885	0.362494	0.457799	0.395758	0.395758
min	16.000000	16.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	28.000000	21.900000	0.000000	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	41.000000	27.800000	0.000000	2.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
75%	53.000000	34.000000	1.000000	3.000000	1.000000	0.000000	1.000000	0.000000	1.000000	0.000000	0.000000
max	65.000000	40.000000	1.000000	4.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000

FIGURE 6 DESCRIPTIVE STATISTICS

3.11 Chapter Summary

This chapter outlined the research methodology employed in the study. It encompassed the research design, target population, sampling procedures, data sources, and analytical techniques utilized to achieve the study's objectives. By employing a quantitative methodology, the research aimed to design a predictive Claims cost model using supervised learning techniques, identify hidden risk factors through supervised learning algorithms, and assess the impact of supervised learning on analysing medical images and physiological signals. The chapter provided a comprehensive overview of each component, ensuring clarity and rigor in the research process.

CHAPTER 4: DATA ANALYSIS, INTERPRETATION AND DISCUSSION

4.0 Introduction

This chapter presents the analysis, interpretation, and discussion of the data collected for the development of a Machine Learning (ML) predictive model for personalized insurance premiums, using CIMAS Medical Aid Society as a case study. The primary aim is to evaluate the potential of supervised ML techniques in accurately predicting individual healthcare costs and mapping these to personalized premium rates, thereby improving fairness, accuracy, and affordability in health insurance pricing in Zimbabwe. The chapter begins by providing an overview of data preprocessing steps undertaken to prepare the structured health and claims data for model training. Subsequently, it details the application of various supervised learning algorithms such as Linear Regression, Decision Trees, Random Forests, and Gradient Boosting Machines, assessing their performance in predicting annual Total claims cost as the target variable for this study. The chapter also includes an exploration of feature importance to understand the key predictors influencing healthcare costs, such as age, chronic illness indicators, hospital visits, and prescription frequency. These insights are crucial for informing personalized premium decisions.

4.1 Data Preprocessing

4.1.1 Correlation Analysis

From the heatmap below, the color of each cell represents the strength and direction of the correlation between variables. The brighter colors indicate stronger correlations whilst darker colors indicate weaker correlation. Therefore, from the above, brighter colors indicate that the respective variables have stronger correlation with utilization factors for instance chronic meds has a strong relationship with num_prescriptions which means that the cost of prescriptions are explained by 65% of chronic medications. The analysis of the correlation matrix reveals several key insights into the relationships among variables in the dataset. Total claim cost exhibits the strongest positive correlations with three variables, the number of emergency room visits (num_er) at 0.72, the heart problem at 67% and the number of prescriptions at 0.33. These suggest that higher healthcare utilization and chronic condition burden are major drivers of overall healthcare expenditure

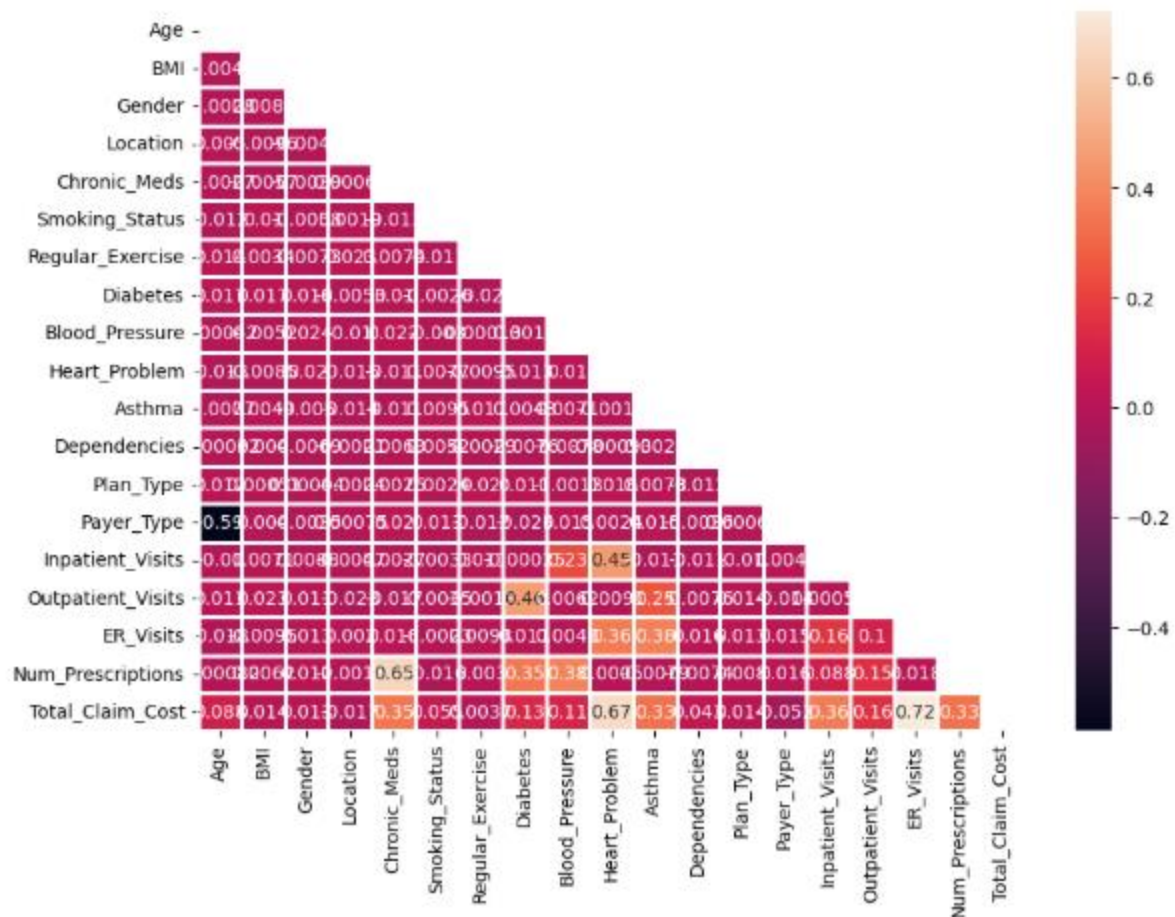


FIGURE 7 CORRELATION MATRIX

4.1.2 Checking for Multicollinearity

The variance inflation factor (VIF) analysis reveals that the constant term (intercept) has a very high VIF value (7.5), which is common and not concerning, as it does not represent a predictor variable. All the individual predictors exhibit VIF values between approximately 1.0 and 3.3, well below the common thresholds (typically 5) used to indicate multicollinearity issues. This suggests that there is no significant multicollinearity among the explanatory variables, meaning the model's predictors are relatively independent and unlikely to distort coefficient estimates due to redundancy.

```

Variance Inflation Factors (VIF):
Feature      VIF
0      const  7.499196
18  Num_Prescriptions  3.225909
5      Chronic_Meds  2.400282
8      Diabetes  1.801109
10     Heart_Problem  1.561976
14     Payer_Type  1.535086
1      Age  1.533665
16  Outpatient_Visits  1.513238
9      Blood_Pressure  1.453653
17     ER_Visits  1.452156
15  Inpatient_Visits  1.438578
11     Asthma  1.352734
7      Regular_Exercise  1.002891
13     Plan_Type  1.001780
3      Gender  1.001733
4      Location  1.001555
2      BMI  1.001135
6      Smoking_Status  1.001054
12     Dependencies  1.000870

⚠ Features with High Multicollinearity (VIF > 5):
Feature      VIF
0      const  7.499196

```

FIGURE 8 CHECKING FOR MULTICOLLINEARITY USING VIF

4.2 Data Transformation

According to Fernando (2021), when building or designing machine learning models, all features should be in numerical form. Some of the attributes in the dataset used for the purposes of the research were categorical data, therefore, there is a need to transform them into an integer or numerical format so as to create a format that the machine learning regression algorithms can understand. There are four attributes in figure that was labelled “object”. Through python the researcher was able to convert the format of the attributes from categorical to numerical. The below figure shows the python code used to change the format of the attributes and the dataset after the data encoding.

```

[8]: label_encoders = {}

for col in cat_cols:
    le = LabelEncoder()
    data[col] = le.fit_transform(data[col])
    label_encoders[col] = le # Store encoder for later use

# Display the first few rows
data[cat_cols].head()

```

```

[8]:   Gender  Location  Plan_Type  Payer_Type
0      1         0         1         1
1      1         4         0         0
2      0         4         2         0
3      1         4         5         1
4      0         1         4         3

```

FIGURE 9 LABEL ENCODING

4.2.1 Checking for Skewness in Data

Checking for skewness is a crucial step in data preprocessing, especially when the modelling technique assumes normality and linearity in the relationships between variables. Skewness refers to the asymmetry of a variable's distribution and is particularly common in health-related datasets, such as insurance claim costs or hospital visits, which tend to be right-skewed due to a small number of individuals incurring extremely high costs. Ignoring skewness can result in biased model coefficients, inefficient estimates, and violations of statistical assumptions such as homoscedasticity and normality of residuals (Kuhn and Johnson, 2019). To mitigate these risks, skewed variables are often log-transformed to achieve more symmetric distributions, improve model accuracy, and support better generalisation (James *et al.*, 2021). Furthermore, in healthcare analytics and actuarial modelling, log-transformations of skewed cost data help in reducing the influence of outliers and ensuring fairness in predictive models (Nguyen *et al.*, 2023).

```
[17]: # Checking skewness
      data.skew(numeric_only=True)

[17]: Age                -0.003792
      BMI                 0.019909
      Gender              0.008401
      Location            -0.025199
      Chronic_Meds         0.902340
      Smoking_Status       1.472376
      Regular_Exercise     0.817213
      Diabetes             1.900550
      Blood_Pressure       0.878728
      Heart_Problem        1.544688
      Asthma               1.960686
      Dependencies         0.016309
      Plan_Type            -0.001999
      Payer_Type           0.948678
      Inpatient_Visits     1.566442
      Outpatient_Visits    1.231267
      ER_Visits            2.081408
      Num_Prescriptions    0.948292
      Total_Claim_Cost     1.778249
      dtype: float64
```

FIGURE 10 CHECKING FOR NORMALITY

The skewness values presented reveal that several variables in the dataset exhibit substantial asymmetry, which may negatively impact the performance and validity of machine learning models. Variables such as total_claim_cost (skewness = 1.77), num_ip (1.56), and num_er (2.08) are highly right-skewed, indicating a heavy concentration of lower values with a long tail on the right. These distributions are typical in health insurance data, where a small proportion of individuals incur significantly higher medical costs or utilize more services (Nguyen *et al.*, 2023). According to Kuhn and Johnson (2019), identifying and correcting

skewness is essential before applying linear regression or machine learning models, as skewed predictors can violate the assumption of normality and distort results. Therefore, power transformer due to its effectiveness in handling both negative and positive values was used to transform highly skewed variables, particularly those with absolute skewness above 1, to improve model accuracy and ensure valid statistical inference (James et al., 2021).

4.3 Feature Selection

Feature selection is the process of isolating the most relevant, non-redundant and consistent features to use in the developing of a model, Kaplan (2022). The reduction in input variables will also lead to the reduction in the size of the dataset for the model which ensures an improvement in the performance of a predictive model and also a reduction in the computational cost of modelling. According to Gupta (2023), the removing of unwanted features from a dataset improves the efficiency and the run time of the model. Feature selection also ensures the reduction in overfitting because less redundant data would mean less possibility of making decisions based on redundant data. In addition, feature selection improves accuracy of the model because less misleading data would mean an improvement in the modelling accuracy and also feature selection ensures a reduction in training time as less data would mean that the machine learning algorithm train faster. Because of this, Kaplan (2022) recommended that feature selection should be done before moving on to the model construction. The researcher used domain knowledge, since one of the objectives is to find the factors that affects health cost from utilization patterns and health factors in health insurance sector. Payer_type, plan_type and location was dropped due to irrelevance and redundant variables.

FEATURE SELECTION AND MODEL CONSTRUCTION

```
# Prepare data (assuming 'total_claim cost' is the target variable)
X = data.drop(columns=['Total_Claim_Cost', 'Payer_Type', 'Plan_Type', 'Location'])
y = data['Total_Claim_Cost']
```

FIGURE 11 FEATURE SELECTION

4.4 Model Training

The researcher started by splitting the research dataset into a training set and test set as this is the norm when using machine learning algorithms to develop models. To evaluate the performance of the predictive models effectively, the dataset was partitioned using a three-step data splitting approach. Initially, the full dataset was divided into training and testing subsets

using a 70:30 ratio, where 70% of the data was allocated for training the models and 15% was reserved for testing and 15% for validating their performance on unseen data. Within the training set, cross-validation was applied, k-fold cross-validation ($k=5$) was used to ensure robust model evaluation and to mitigate the risk of overfitting. This method allowed each subset of the training data to be used both for training and validation, thus enhancing the generalizability of the models before their final evaluation on the hold-out test set.

```
[20]: from sklearn.model_selection import train_test_split

# First split: training set and temporary set (60% train, 40% temp)
X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.3, random_state=42)

# Second split: validation and test sets (split temp into 50/50)
X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.5, random_state=42)

# Confirm the sizes
print(f"Training set size: {X_train.shape[0]}")
print(f"Validation set size: {X_val.shape[0]}")
print(f"Test set size: {X_test.shape[0]}")

Training set size: 7000
Validation set size: 1500
Test set size: 1500
```

FIGURE 12 SPITTING INTO TRAIN AND TESTING

4.4.1 Cross-Validation and Hyperparameter Tuning

To enhance the accuracy and efficiency of the models, hyperparameter tuning was applied across all selected algorithms, including Linear Regression, Random Forest, Artificial Neural Networks (ANN), XGBoost, and Gradient Boosting. For tree-based models such as Random Forest, XGBoost, and Gradient Boosting, the tuning process targeted key parameters such as the number of estimators, maximum tree depth, and learning rate. Optimization was achieved using Grid Search and Randomized Search, which systematically explore combinations of hyperparameters to identify the most effective settings. The ANN model was tuned by adjusting the number of hidden layers, neurons per layer, activation functions, and epochs through a combination of manual tuning and Keras GridSearchCV wrapper. Cross-validation was incorporated during tuning to ensure the models generalized well and to select the best parameter combinations based on validation performance.

```
RandomForestRegressor performance:
  Train R²      : 0.9940
  Test R²       : 0.9527
  CV R² (5-Fold) : 0.9559
Best hyperparameters: {'n_estimators': 150}
Best CV R² (grid) : 0.9565
```

```
XGBRegressor (default) performance:
  Train R²      : 0.9875
  Test R²       : 0.9497
  CV R² (5-Fold) : 0.9551
Best hyperparameters: {'gamma': 0, 'max_depth': 5, 'n_estimators': 40}
Best CV R² (grid): 0.9593
XGBRegressor (tuned) performance:
  Train R²      : 0.9711
  Test R²       : 0.9526
  CV R² (5-Fold) : 0.9593
```

FIGURE 13 HYPERPARAMETERS TUNNING

4.5 Application of Machine Learning Algorithms

After the above, the researcher trained the regression models using different regression machine learning algorithms. The models are available in the sklearn python library; therefore, this made the training process much easier for the researcher. various supervised machine learning algorithms were applied to train and test models for predicting healthcare costs. The models included Linear Regression, Random Forest, Gradient Boosting, XGBoost, and an Artificial Neural Network (ANN). After preprocessing and transforming the dataset, the models were trained on the training set and evaluated on the test set to assess their predictive capabilities. Each model captured different patterns within the data. Linear Regression provided a simple baseline, while tree-based models and ANN leveraged their complexity to model non-linear relationships. Performance was evaluated using metrics such as R^2 , RMSE, and MSE, with cross-validation used to ensure consistency and generalization of the results. The best performing model was chosen which is the

Final Model Comparison Table

	Model	Train R ²	Test R ²	Train RMSE	Test RMSE	Train MSE
0	Linear Regression	0.8989	0.8887	0.32	0.33	0.10
1	Random Forest	0.9945	0.9563	0.07	0.21	0.01
2	Gradient Boosting	0.9617	0.9505	0.20	0.22	0.04
3	XGBRegressor	0.9913	0.9527	0.09	0.22	0.01
4	ANN (Keras)	0.9740	0.9531	0.16	0.22	0.03

	Test MSE	CV R ²
0	0.11	0.8937
1	0.04	0.9589
2	0.05	0.9552
3	0.05	0.9572
4	0.05	0.9563

The model comparison results demonstrate that all five machine learning models (Linear Regression, Random Forest, Gradient Boosting, XGBoost, and Artificial Neural Networks) performed relatively well in predicting the target variable, with R² values above 0.88 on the test set. Among them, Random Forest emerged as the top-performing model, achieving the highest Test R² of 0.9563 and the lowest Test RMSE of 0.21, indicating excellent predictive accuracy and generalization capability. It also recorded the lowest Test MSE (0.04) and highest Cross-Validation R² (0.9589), confirming consistent performance across validation folds.

Although the ANN, XGBoost, and Gradient Boosting models also performed well with similar metrics (Test R² around 0.95 and RMSE of 0.22), Random Forest slightly outperformed them in most key areas. Linear Regression, while much simpler, showed comparatively lower performance with a Test R² of 0.8887 and a higher RMSE of 0.33, indicating it may not capture the complexity of the data as effectively.

4.5.1 Model Justification

The Random Forest Regressor is selected as the best model due to its outstanding balance between predictive power and generalization. It achieved the best overall test metrics, indicating both high accuracy and robustness to overfitting. Unlike ANN and XGBoost, Random Forest is also more interpretable and easier to tune, which is beneficial in health insurance applications where transparency and explanation of predictions are important. Furthermore, its low variance across folds in cross-validation supports its reliability for deployment in real-world scenarios.

4.5.2 Evaluation Metrics

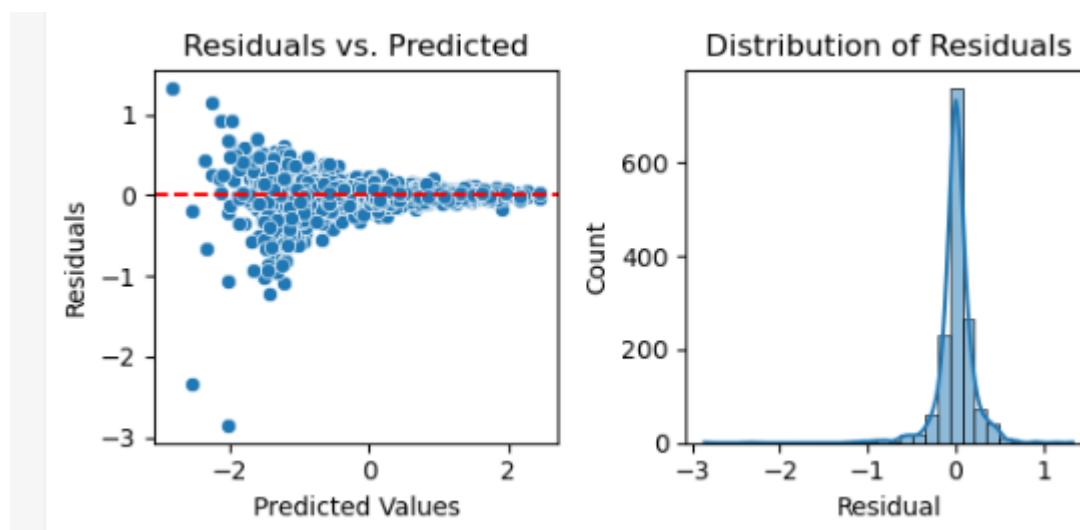
To assess the effectiveness of the regression models, three key performance metrics were employed: R-squared (R^2), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

R^2 , also known as the coefficient of determination, indicates the proportion of the variance in the dependent variable that can be explained by the independent variables. Its values range from 0 to 1, where values closer to 1 suggest a stronger model fit. Generally, an R^2 value above 0.9 is considered excellent, between 0.8 and 0.9 is regarded as good, while values below 0.7 may suggest a weak model.

MSE evaluates model accuracy by averaging the squared differences between predicted and actual values. Since it squares the errors, MSE is highly sensitive to outliers and large errors, making it a suitable measure for identifying substantial prediction deviations. A lower MSE signifies higher prediction accuracy, although what qualifies as "low" depends on the scale of the dependent variable.

RMSE, the square root of the MSE, converts the error back into the same units as the predicted variable, enhancing interpretability. Like MSE, a lower RMSE indicates better performance and is especially useful for comparing how closely different models' predictions align with actual values (Chicco, Warrens & Jurman, 2021).

In this study, models with RMSEs between 0.21 and 0.33 and R^2 values above 0.88 demonstrate strong predictive power, with the Random Forest model achieving the most favorable balance across all three metrics.



The residual analysis presented in the figure offers a critical evaluation of the performance of the Random Forest regression model. The residuals versus predicted values plot reveals that the residuals are predominantly centered around zero, indicating that the model does not systematically overestimate or underestimate healthcare costs. However, the presence of a funnel-shaped pattern, where residual variance decreases as predicted values increase, suggests heteroscedasticity in the model's predictions. This implies that the Random Forest model demonstrates greater accuracy for higher-cost predictions while exhibiting more variability and less precision for lower-cost cases. Despite this, the histogram of residuals shows an approximately normal distribution centered around zero, which indicates that the errors are symmetrically distributed and the model is generally well-calibrated. A slight long tail is observed in the distribution, which can be attributed to the structure of the input data specifically, the predominance of binary (yes/no) predictors such as chronic conditions and lifestyle indicators.

4.5.3 Predicted Cost and Residuals

Index	Actual (Original)	Predicted (Original)	Residual (Original)
2697	2,949.81	2,074.23	875.58
6871	7,867.10	6,696.50	1,170.60
3487	1,701.01	1,299.49	401.52
92	7,108.70	6,767.95	340.75
9537	7,543.58	7,684.10	-140.52
3205	16,051.75	14,322.53	1,729.22
6641	12,293.62	12,140.78	152.84
8909	34,384.81	33,327.52	1,057.28
2884	1,869.19	2,347.35	-478.16
7173	18,293.89	20,040.72	-1,746.83

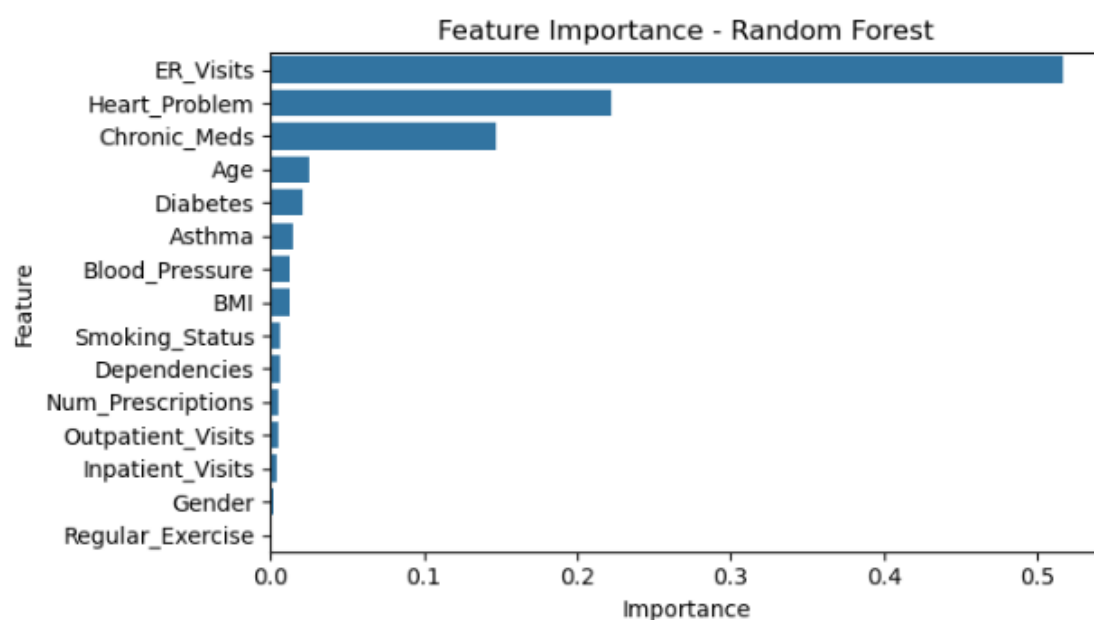
TABLE 3 PREDICTED COST AND RESIDUALS

The model demonstrates strong performance in predicting the Total_Claim_Cost variable (which was transformed and later reversed to its original form). For high-cost cases (e.g., Index 8909 with an actual cost of \$34,384), the predicted cost (\$33,327) is remarkably close, with a small residual of \$1,057. This shows that the model handles larger claim costs well, which is essential in insurance applications where high-cost outliers can significantly affect profit/loss.

For low-cost records like Index 3487 (Actual: \$1,701, Predicted: \$1,299), the prediction slightly underestimates but remains within a reasonable margin. The residuals are not drastically large, suggesting consistency.

Residuals in the data are generally small (both positive and negative), indicating low prediction error. Most are below \$1,000, suggesting that the model doesn't systematically under- or over-predict. The model doesn't show signs of bias in one direction. For instance, Index 9537 has a slight over-prediction, while Index 2884 is slightly under-predicted. This balance is a good sign of generalization.

4.5.4 Feature Importance



The feature importance chart from the Random Forest model visually conveys how much each predictor contributes to the model's ability to forecast healthcare costs. Here's an academic-style interpretation:

The Random Forest regression model's most influential factors in predicting individual healthcare costs. The figure indicates that ER_Visits (Emergency Room Visits) is by far the most significant predictor, contributing to over 50% of the model's decision-making power. This highlights that frequent emergency room utilization is strongly correlated with higher healthcare expenditures.

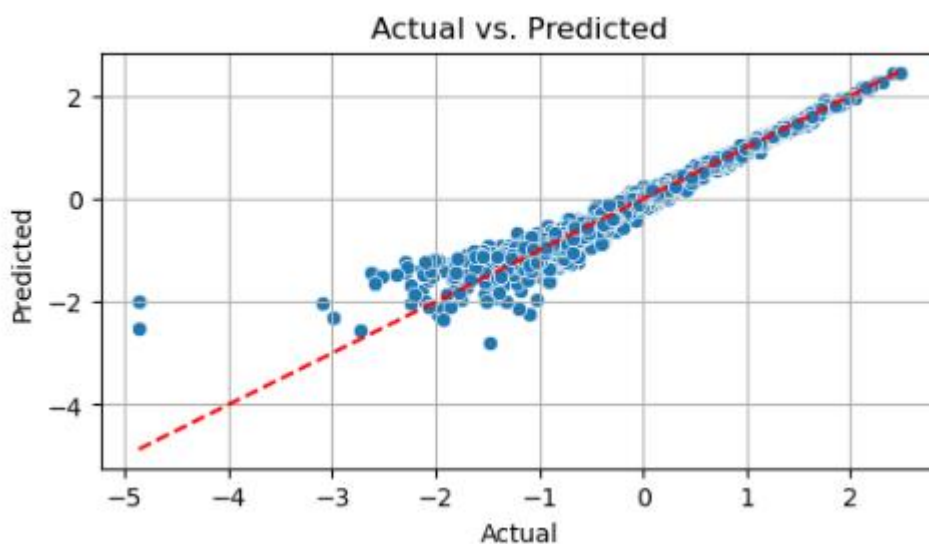
Following ER_Visits, Heart_Problem and Chronic_Meds (indicating the presence of chronic conditions requiring regular medication) are the next most important features, suggesting that

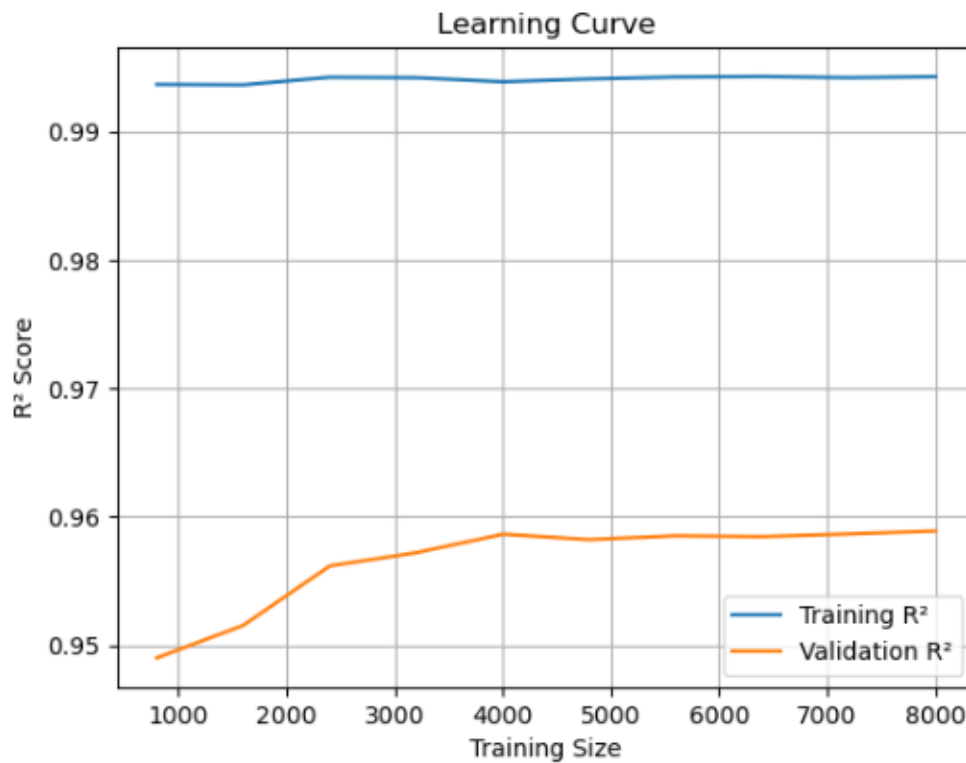
chronic illness and associated medication costs substantially affect overall healthcare spending. Age also appears with moderate importance, reinforcing existing literature that aging correlates with increasing health risks and utilization.

Other features such as Diabetes, Asthma, and Blood_Pressure have lower but noticeable importance, consistent with their roles in managing chronic disease risk. Interestingly, variables traditionally thought to influence cost, such as BMI, Smoking_Status, and Inpatient_Visits, exhibit relatively low importance in this specific model. This may suggest that their individual predictive power is limited in the presence of more direct indicators of healthcare utilization like ER visits and chronic disease markers.

Low-importance features such as Gender and Regular_Exercise may still hold clinical significance but are overshadowed in this predictive context by more acute cost drivers. Overall, the model highlights that acute utilization events (e.g., ER visits) and chronic disease indicators are central to forecasting healthcare costs, which can be instrumental in guiding personalized premium assignments and resource allocation in health insurance frameworks.

Scatter Plot





4.6 Premium Mapping

The personalized premium mapping presented below demonstrates the application of a machine learning-based health cost prediction model to optimize insurance plan assignment based on individual health risk profiles. By comparing the predicted annual healthcare cost against predefined benefit thresholds, members were assigned to the most appropriate insurance plan using a "tied option" strategy. This ensures that individuals are placed in plans whose benefit limits closely match their expected healthcare expenditure, thereby promoting actuarial fairness and potentially enhancing affordability and sustainability for both the insurer and the insured.

The analysis reveals both matches and mismatches between the model-assigned plan and the actual plan the member was enrolled in. For example, a member with a predicted cost of \$2,093.06 was correctly assigned to the Health Save plan, which is appropriate given the low risk. However, other instances highlight significant mismatches. One individual with a predicted cost of \$1,215.44 was enrolled in the Secure Premium plan, which provides significantly higher coverage than needed, leading to a plan mismatch. Similarly, a member with a predicted cost of \$33,319.97 was only on the Secure Primary plan, despite their risk justifying a Secure Premium assignment, indicating under coverage.

These mismatches emphasize the limitations of traditional plan selection processes, which may rely on static demographic categories or voluntary selection rather than data-driven health risk assessments. By integrating machine learning predictions into the premium assignment process, insurers can align benefit structures with actual risk more precisely, supporting better resource allocation, cost containment, and personalized service delivery in health insurance.

Index	Actual Cost	Predicted Cost	Residual	Assigned Plan	Actual Plan	Comparison
2697	2949.81	2093.06	856.76	Health Save	Health Save	Match
6871	7867.10	6665.42	1201.68	Secure Essential	Secure Primary	Mismatch
3487	1701.01	1215.44	485.57	Health Save	Secure Premium	Mismatch
92	7108.70	6779.51	329.19	Secure Essential	Secure Premium	Mismatch
9537	7543.58	7726.47	-182.89	Secure Essential	Secure Primary	Mismatch
3205	16051.75	14349.25	1702.51	Secure Essential	Secure Essential	Match
6641	12293.62	12217.93	75.68	Secure Essential	Secure Premium	Mismatch
8909	34384.81	33319.97	1064.84	Secure Premium	Secure Primary	Mismatch
2884	1869.19	2317.19	-448.00	Health Save	Secure Premium	Mismatch
7173	18293.89	20013.66	-1719.76	Secure Private	Secure Premium	Mismatch

TABLE 4 ACTUAL AND PREDICTED COST

4.7 Model Saving and Deployment

Once the machine learning model was trained and evaluated, the next critical step is to prepare it for real-world application. Saving the trained model ensures it can be reused without needing to retrain, saving time and computational resources. In this study, the best-performing model Random Forest Regression was serialized using Python's joblib libraries.

The model can be integrated into CIMAS's health insurance system using a web-based API or a dashboard interface, where member data is input and the system predicts healthcare cost or suggests a personalized premium. For deployment, platforms like Modelbit could be used for interactive access. Future extensions could involve embedding the model into a larger decision support tool, linked with CIMAS's claims and health records database for real-time premium recommendations.

To ensure reliability, model versioning, regular updates with new data, and performance monitoring would be necessary. Furthermore, any deployment would need to comply with data privacy laws and ethical AI practices in Zimbabwe's healthcare sector.

4.8 Chapter summary

This chapter presented the implementation and evaluation of various supervised machine learning algorithms designed to predict personalized healthcare costs using structured data. After rigorous preprocessing, training, and cross-validation, the Random Forest Regression model emerged as the most effective, achieving high predictive accuracy with an R^2 of 0.9563 and a low RMSE.

Feature importance analysis highlighted that emergency room visits, chronic disease presence, and prescription usage were the most significant contributors to healthcare costs. These insights not only validate the model's robustness but also align with real-world cost drivers in health insurance.

The chapter concluded with a discussion on how the model can be saved and deployed for practical use in the health insurance sector, particularly for CIMAS. The model's predictive capabilities open opportunities for data-driven decision-making in premium pricing and resource allocation.

Chapter 5: Summary, Conclusions, and Recommendations

5.0 Summary

This study explored the application of machine learning (ML) models in personalizing medical insurance premiums based on health risk factors and healthcare utilization patterns. The study focused on the context of CIMAS Medical Aid Society in Zimbabwe, a leading healthcare insurer operating in a dynamic and resource-constrained environment. It sought to address the limitations of conventional actuarial methods that primarily rely on demographic data such as age and gender for determining insurance premiums, often resulting in generalized and potentially inequitable pricing structures.

The research commenced with a detailed contextual and theoretical background, emphasizing the global rise in healthcare costs and the growing demand for data-driven premium pricing models that enhance equity, accuracy, and financial sustainability. Through a comprehensive review of related empirical literature, the study identified a range of supervised machine learning algorithms used internationally for health cost prediction, including Random Forest Regression, Gradient Boosting Machines (GBM), XGBoost, Artificial Neural Networks (ANN), and Linear Regression.

Following model development and testing, the Random Forest Regression model emerged as the most effective in predicting annual healthcare expenditure, outperforming its counterparts in accuracy, robustness, and generalizability. The model leveraged a combination of demographic, clinical, and utilization-based predictors to assess individual health risk and estimate likely annual costs, forming the basis for personalized premium recommendations.

The analysis presented in Chapter 4 revealed that healthcare costs are predominantly influenced by acute and chronic care factors such as emergency room visits (ER_Visits), chronic medication usage (Chronic_Meds), and cardiovascular conditions (Heart_Problem). These findings corroborated prior research, reinforcing the centrality of chronic disease management in healthcare expenditure forecasting. The study also exposed mismatches between existing plan allocations and recommended premium tiers, highlighting the inefficiencies of non-personalized premium assignment. The results underscore the transformative potential of AI-enabled health risk models in achieving fairer, more efficient, and more sustainable insurance pricing mechanisms in Zimbabwe.

5.1 Chapter summary

The study successfully designed, trained, and validated a Random Forest regression model capable of predicting annual healthcare costs with a high degree of accuracy ($R^2 = 0.9563$). This confirms the efficacy of machine learning in modeling complex, non-linear relationships between health-related variables and medical expenditures. The model demonstrated strong predictive performance even in the presence of multiple interacting variables, a task where traditional statistical models often fall short.

The ability to utilize rich health data spanning demographics, chronic disease indicators, and historical utilization patterns allowed for individualized risk assessment. As such, health insurers like CIMAS can use these models to design tailored insurance plans, optimize pricing, and reduce adverse selection, ultimately enhancing both actuarial fairness and member satisfaction.

The feature importance analysis confirmed that acute care utilization metrics (e.g., emergency room visits) and chronic condition indicators (e.g., long-term medication, heart problems) were the strongest predictors of medical cost. In contrast, lifestyle-related variables such as BMI and smoking status, though relevant, were found to have relatively weaker predictive influence in the model. This suggests that actual service utilization is more directly tied to short- to medium-term healthcare costs than potential or latent health risks.

These findings highlight a crucial insight: while preventive health indicators are vital from a public health perspective, insurers must prioritize utilization-based metrics when pricing risk in the short term. Therefore, real-time access to healthcare usage data could significantly enhance premium forecasting accuracy.

Among the ML models tested, Random Forest Regression outperformed Linear Regression, Gradient Boosting, XGBoost, and Artificial Neural Networks in terms of predictive accuracy, feature robustness, and handling of multi-collinearity. This finding reinforces the growing body of evidence that ensemble-based learning methods, which combine the strengths of multiple decision trees, are particularly well-suited for health cost modeling tasks.

The result suggests that insurers seeking to operationalize machine learning for premium personalization should consider starting with ensemble models due to their adaptability, interpretability, and superior predictive capabilities in heterogeneous datasets.

Objective 1: To design a machine learning predictive model for personalized insurance premiums.

The research achieved this objective by developing a data-driven predictive model using supervised machine learning algorithms. The model leveraged structured health and claims data from CIMAS to forecast annual individual healthcare costs, which were then used to recommend personalized premiums. The application of the CRISP-DM methodology facilitated a systematic modelling process from data understanding to evaluation. The best-performing model, Random Forest Regression, demonstrated a high degree of accuracy ($R^2 = 0.9563$, RMSE = 0.21), indicating that machine learning can provide more reliable predictions than traditional actuarial methods. This highlights the practical viability of implementing ML-powered premium pricing frameworks in Zimbabwe's insurance sector.

Objective 2: To investigate the relationship between individual health factors, healthcare utilization patterns, and healthcare costs in determining personalized premiums.

Through correlation analysis, feature importance metrics, and model evaluation, the study identified a set of critical predictors for healthcare expenditure. These include the number of emergency room visits, number of prescriptions, presence of heart problems, chronic medication use, and comorbidity index. These variables showed a strong positive correlation with total claim cost. Interestingly, some commonly assumed cost drivers such as BMI, smoking status, and regular exercise had weaker influence in the model. This challenges traditional assumptions and emphasizes the importance of real utilization and chronic disease indicators over static demographic data in cost estimation. These findings reinforce the need for insurers to re-evaluate the factors they consider in pricing models, ensuring they reflect actual risk and cost behaviors.

Objective 3: To evaluate and compare the effectiveness of advanced machine learning algorithms in predicting personalized medical insurance premiums.

The study compared several algorithms: Linear Regression, Gradient Boosting, XGBoost, Artificial Neural Networks, and Random Forests. While all models achieved reasonable performance, ensemble-based models like Random Forest and XGBoost delivered superior predictive capabilities. The Random Forest model offered the best balance between accuracy, interpretability, and generalizability, validating its use as a practical tool for insurance companies. This suggests that ML algorithm selection matters significantly, and deploying

complex but explainable models can lead to meaningful improvements in how premiums are structured.

5.2 Recommendations

It is recommended that CIMAS and other health insurers in Zimbabwe formally adopt AI-driven premium personalization frameworks. By utilizing predictive models such as the one designed in this study, insurers can move away from blanket pricing schemes toward a risk-adjusted pricing strategy that aligns premiums with individual health profiles. This approach will not only enhance affordability and fairness for members but also help insurers mitigate financial losses due to underpricing high-risk individuals. By training these models on historical claims and health data, insurers can shift from one-size-fits-all pricing to precision pricing. This transition should begin with a pilot program targeting specific member categories to test feasibility and gather feedback.

To promote transparency and trust, machine learning models used in premium pricing should be accompanied by explainability frameworks such as SHAP (SHapley Additive exPlanations) or LIME. These tools allow insurers to justify premium decisions to regulators and members by showing the individual-level contribution of each feature to the predicted cost. By fostering explainability, insurers can address concerns around algorithmic bias and enhance the ethical governance of AI applications in healthcare finance.

To further enhance the predictive accuracy and responsiveness of AI-based premium models, future systems should incorporate real-time health monitoring data (e.g., wearable device data, fitness tracking, sleep patterns) alongside Electronic Health Records (EHRs) and insurance claims. This multi-source integration will allow insurers to develop a dynamic, adaptive risk assessment system that evolves with changes in an individual's health status, enabling ongoing premium adjustments and personalized interventions.

The model revealed clear mismatches between risk profiles and selected plans: some low-risk members are in high-cost plans, and vice versa. CIMAS should use the model to guide members into risk-aligned plans, optimizing utilization and financial sustainability. This can be achieved through tiered premium mapping, where risk scores are matched to plan categories, ensuring consistency between expected cost and plan coverage.

Given the transformative potential of AI in health insurance, policymakers must develop regulatory frameworks that ensure ethical deployment. This includes establishing data privacy standards, fairness benchmarks, and transparency mandates for AI models in pricing decisions.

Guidelines should also address the risks of discriminatory pricing, particularly against vulnerable populations such as those with pre-existing conditions, and promote inclusive access to affordable healthcare plans.

5.3 Areas for Future Research

Future research should explore whether advanced deep learning architectures, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models, can provide improved predictive accuracy by capturing temporal patterns in healthcare utilization. Investigate the use of fairness-aware machine learning techniques, including re-weighting, adversarial debiasing, and differential privacy, to ensure that premium pricing does not systematically disadvantage any particular group based on gender, age, or pre-existing medical conditions.

Researchers should examine the operational and predictive implications of integrating IoT-based health tracking systems into insurance risk models. This could include studies on how wearable-generated health metrics impact cost prediction and preventive care planning. Conducting comparative studies of AI-driven premium pricing across developed and developing healthcare systems, focusing on scalability, cultural receptiveness, infrastructural readiness, and regulatory frameworks. Such analyses can help tailor AI adoption strategies to diverse health financing environments.

5.4 Chapter summary

In conclusion, this study demonstrates that machine learning models particularly ensemble techniques like Random Forest Regression have the potential to revolutionize premium pricing strategies in the health insurance sector. By leveraging structured healthcare data and predictive analytics, insurers can achieve a more equitable allocation of premiums, reduce systemic inefficiencies, and enhance both financial performance and customer satisfaction.

As Zimbabwe and other developing nations continue to navigate healthcare reforms, the integration of AI-powered personalized pricing models will be instrumental in building a resilient, data-driven, and inclusive insurance ecosystem. The future of medical insurance lies in personalization, transparency, and continuous innovation, where predictive models are not just tools for forecasting costs, but strategic assets for driving health equity and access.

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APENDIX 1

MODEL CODES SNIPPETS